Probability & Statistics
Open-Book Binder

Quiz 1 & Midterm (UTS) · Sessions 1–12 · Devore, *Probability and Statistics for Engineering and the Sciences*, 9E

English with Indonesian terms in *(kurung / brackets)* · Calculator: Casio fx-991ID (DIST mode)

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How to use this binder in the exam

1. **Translate** the question with Tab G. The keywords that decide the method are *paling sedikit, paling banyak, tidak lebih dari, jika diketahui, tanpa pengembalian, rata-rata ... per*.
2. **Classify** it with Tab I (index) or Tab D (flowchart). This tells you the distribution/rule *and* the page.
3. **Compute** with the worked example on that page as a template. Use DIST mode to check (Tab C), but **write the formula and the substitution** on your paper. The class boards show the professor expects working, not just

answers.

4. **Sanity-check**: a probability must be between 0 and 1, a variance must be ≥ 0, a pdf must integrate to 1, and probabilities in a table must sum to 1.

> ✔ Every number in the worked examples and the answer key was recomputed by computer. Where a board photo had a slip, the corrected value is used and the slip is listed on the Traps and errata page (Tab X).

Sources: the lecture decks (Overview & Descriptive Statistics, Basics Probability, Discrete RVs, Continuous RVs; slides by Anom Yudistira, based on Devore 9E), the readings in the course repo (Purdue MA416 slides, Univ. of Sydney MLC notes, probabilitycourse.com §4.1–4.2), whiteboard photos from class, and your OneNote notes (Probstat 5, 9, 11).

G · Glossary: Indonesian question words → meaning → math

The most dangerous words are the ones that change **≤ vs <** or **which side** of the distribution you need. They are marked ★.

Inequalities

Indonesian	English	Math
★ paling sedikit / minimal / sekurang-kurangnya	at least	$X \geq k$
★ paling banyak / maksimal / sebanyak-banyaknya	at most	$X \leq k$
★ tidak lebih dari	not more than	$X \leq k$
★ tidak kurang dari	not less than	$X \geq k$
lebih dari	more than	$X > k$
kurang dari	less than	$X < k$
tepat / sebanyak / ada (k)	exactly	$X = k$
antara a dan b	between a and b	$a \leq X \leq b$*
ada 2–4 / dari 2 sampai 4	from 2 to 4	$2 \leq X \leq 4$
sedikitnya satu / minimal satu	at least one	$1 - \mathrm{P}(X = 0)$
tidak ada / tidak satu pun	none	$X = 0$
semua	all	$X = n$

*For a **continuous** X, ≤ and < give the same answer. For a **discrete** X (Binomial, Poisson) they do not. If "antara" is ambiguous for a discrete X, write down which you assumed, e.g. "diasumsikan inklusif" (assumed inclusive).

Instructions

Hitunglah / Tentukan / Carilah	Calculate / Determine / Find
Buktikan bahwa ... merupakan pdf	Prove that ... is a valid pdf (check f ≥ 0 and ∫ = 1)
Lengkapi tabel	Complete the table
Diketahui	Given / It is known that
Jika / apabila	If
Jelaskan / Interpretasikan	Explain / Interpret
Gambarkan	Draw / sketch
secara acak	at random
Berapa peluang ...	What is the probability ...
Berapa banyak cara ...	How many ways ... (counting)

Probability words

peluang / probabilitas	probability
kejadian	event
ruang sampel	sample space S
percobaan	experiment
saling lepas / saling asing	mutually exclusive ($A \cap B = \emptyset$)
saling bebas / independen	independent ($\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$)
peluang bersyarat	conditional probability $\mathrm{P}(A \mid B)$
★ jika diketahui / dengan syarat / diberikan	given that → the condition goes **after** the bar
gabungan (∪) / irisan (∩)	union / intersection
komplemen / bukan	complement A^c
peluang gabungan (joint)	joint probability $\mathrm{P}(A \cap B)$
peluang marginal	marginal probability (row/column total ÷ grand total)
himpunan kosong	empty set ∅
permutasi / kombinasi	permutation (order matters) / combination
★ tanpa pengembalian	without replacement → Hypergeometric
dengan pengembalian	with replacement → Binomial
dadu / koin / kartu / bola / kotak / guci	die / coin / card / ball / box / urn
mata dadu / selisih / jumlah	die face / difference / sum
cacat / rusak / baik	defective / broken / good
lulus / tidak lulus (TL) / mengulang	pass / fail / retake

Statistics & distribution words

variabel random / acak (diskrit / kontinyu)	random variable (discrete / continuous)
fungsi peluang (pmf / pdf)	probability function (pmf / pdf)
fungsi kumulatif (cdf)	cumulative distribution function F(x)
rata-rata / nilai harapan	mean / expected value $\mu = E(X)$
varians / ragam	variance σ^2
simpangan baku / standar deviasi	standard deviation σ
median / modus / kuartil / persentil	median / mode / quartile / percentile
jangkauan	range
pencilan	outlier
populasi / sampel	population (N, μ, σ) / sample ($n, \bar{x}, s$)
★ laju / rata-rata kejadian per (jam, hari, halaman)	rate λ → Poisson (count) or Exponential (time)
waktu tunggu / umur / masa pakai	waiting time / lifetime → Exponential
distribusi normal baku	standard normal $Z \sim N(0,1)$
seragam	uniform
sukses / gagal	success / failure (Binomial)
ketebalan / berat bersih / tinggi badan / suhu	thickness / net weight / height / temperature

⚠ **TRAP** Indonesian decimals use a **comma** and thousands use a **dot**: "0,4" = 0.4 and "15.015" = 15015 (the class normal-distribution example). Read every number twice.

I · Problem-type index: "I see … → I do … → page"

Look up the **phrase or setup** in the question, not the chapter name.

The question says / shows …	Method	Page
1 · Data sets (descriptive statistics)		
a list of numbers: "rata-rata, median, modus"	$\bar{x} = \sum x/n$; median at position $0.5(n+1)$	p.10
"varians / simpangan baku" of data	$s^2 = \dfrac{\sum x^2 - (\sum x)^2/n}{n-1}$; STAT mode check	p.10
"kuartil, IQR, box plot, pencilan (outlier), five-number summary"	Q1 at $0.25(n+1)$, Q3 at $0.75(n+1)$; fences $Q_1 - 1.5\,IQR,\ Q_3 + 1.5\,IQR$	p.11
"at least what proportion within k std dev" (no shape given)	Chebyshev $1 - 1/k^2$	p.10
"mound-shaped / bell-shaped", 68–95–99.7	Empirical rule	p.10
qualitative vs quantitative, discrete vs continuous, skewness, stem-and-leaf	Definitions + shape rules (mean vs median)	p.10
2 · Events and probability		
a two-way table of counts (gender × lulus, age × insurance …)	joint = cell/total; marginal = row or column total/total; conditional = cell/row-or-column total	p.13
"A atau B" (A or B), "A dan B" (A and B)	$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$	p.12
"jika diketahui / given that"	$\mathrm{P}(A \mid B) = \mathrm{P}(A \cap B)/\mathrm{P}(B)$	p.14
"apakah A dan B saling bebas?" (are A and B independent?)	check $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$ *or* $\mathrm{P}(A \mid B) = \mathrm{P}(A)$	p.14
"saling lepas?" (mutually exclusive?)	check $\mathrm{P}(A \cap B) = 0$. **Not** the same as independent!	p.14
"at least one" / "sedikitnya satu"	complement: $1 - \mathrm{P}(\text{none})$	p.12
several machines/sources with % share and % defect, "P(defective)"	Law of total probability (tree)	p.14
… "found defective; P(it came from machine B)?"	Bayes' rule	p.14
"berapa banyak cara" (how many ways), arrange, choose, committee, password	mn rule / P_r^n / C_r^n	p.12
drawing items from a box, equally likely outcomes	$\mathrm{P}(A) = n(A)/N$ with combinations	p.12
3 · Discrete random variables		
a table of x and P(x), possibly with an unknown k	$\sum p(x) = 1$ to find k; then E, Var	p.15
"tentukan distribusi peluang X" (dice, coins, selisih/jumlah dadu)	list the outcomes, count them, build the table	p.15
E(X), Var(X) from a table, E(aX+b)	$\sum xp(x)$, $\sum x^2p(x) - \mu^2$, $a\mu + b$, $a^2\sigma^2$	p.15
n independent trials, same p, count successes	**Binomial** B(n, p)	p.16
"rata-rata λ kejadian per (jam/hari/halaman/km)", count events	**Poisson**(λ); rescale λ to the asked interval	p.16
draw n from N *without replacement*, M "successes" in the lot	**Hypergeometric**	p.17

The question says / shows …	Method	Page
"until the first success" / "the r-th success"	Geometric / Negative Binomial	p.17
Binomial with large n (np ≥ 5 and nq ≥ 5)	Normal approximation + continuity correction ±0.5	p.21
4–6 · Continuous random variables		
"Buktikan f(x) merupakan pdf" (prove it is a pdf) / "find c (konstanta)"	$f \geq 0$ and $\int f = 1$	p.18
f(x) given, "P(a ≤ X ≤ b)"	$\int_a^b f(x)\,dx$ (calculator ∫ key)	p.18
"fungsi kumulatif F(x)", "cdf"	$F(x) = \int_{-\infty}^{x} f(t)\,dt$, piecewise	p.18
"median / persentil ke-p / η(p)" for a pdf	solve $F(\eta) = p$	p.18
mean / variance of a pdf	$\int xf$, $\int x^2 f - \mu^2$	p.19
"uniformly distributed / seragam" on [A, B]	Uniform: $(x - A)/(B - A)$	p.19
the worked class example f(x) = 0,075x + 0,2 on [3, 5]	full solution a–f	p.19
"berdistribusi normal" N(μ, σ²), "P(X < x)"	$z = (x - \mu)/\sigma$ → z-table / Normal CD	p.20
"top 5% / only 1% exceed / find x such that P = …"	work backwards: $x = \mu + z\sigma$	p.21
"waktu tunggu / umur / lifetime", "rata-rata waktu 90 menit"	**Exponential**, λ = 1/mean	p.22
"given it has already lasted t …"	memoryless: $\mathrm{P}(X > t + s \mid X > t) = e^{-\lambda s}$	p.22
"Y = g(X), find the pdf/cdf of Y"	cdf method: $F_Y(y) = \mathrm{P}(g(X) \leq y)$, then differentiate	p.22
Tools		
calculator: DIST, ∫, STAT, nCr	keystrokes + check values	p.8
z-table / binomial table / Poisson table	Tab T	p.24
"something looks wrong on the board/notes"	Traps and errata	p.23

D · Distribution picker

Is X a COUNT (0, 1, 2, ...) or a MEASUREMENT (time, weight, length, score)?

COUNT = discrete

Is there a fixed number of trials n?

- **Yes, n trials, each success/failure, same p, independent** (with replacement, or a huge population)
  → **Binomial B(n, p)** · p.16
- **Yes, but drawn without replacement from a small lot** of N items, M of them "successes"
  → **Hypergeometric** · p.17
  Rule of thumb: if n/N ≤ 0.05 the Binomial is a fine approximation.

No fixed n, counting events in an interval of time/space at an average rate
→ **Poisson(λ)** · p.16

- λ must match the interval asked: 4 per hour → λ = 2 per 30 min.

Counting trials until the 1st (or r-th) success
→ **Geometric / Negative Binomial** · p.17

A small custom experiment (dice, coins, a given table)
→ build the pmf table yourself · p.15

MEASUREMENT = continuous

A formula f(x) is given
→ general pdf: integrate · p.18

"Equally likely anywhere between A and B" / "seragam"
→ **Uniform[A, B]** · p.19

Bell-shaped, given mean and variance (or SD); heights, weights, thicknesses, scores
→ **Normal N(μ, σ²)** · p.20

Waiting time / lifetime / time between events, x ≥ 0, "rata-rata waktu ..."
→ **Exponential(λ)**, λ = 1/mean · p.22

Binomial with large n (np ≥ 5, n(1-p) ≥ 5) and no table/calculator support
→ **Normal approximation** with continuity correction · p.21

Master formula table

Distribution	pmf / pdf	Range	Mean E(X)	Variance V(X)	Calculator
Binomial B(n, p)	$\binom{n}{x}p^x(1-p)^{n-x}$	0,1,...,n	np	$np(1-p)$	Binomial PD / CD
Poisson(λ)	$\frac{e^{-\lambda}\lambda^x}{x!}$	0,1,2,...	λ	λ	Poisson PD / CD
Hypergeometric h(x; n, M, N)	$\frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$	max(0, n-N+M) ... min(n, M)	$n\frac{M}{N}$	$n\frac{M}{N}\left(1-\frac{M}{N}\right)\frac{N-n}{N-1}$	nCr
Geometric (trial # of 1st success)	$(1-p)^{x-1}p$	1,2,...	$1/p$	$(1-p)/p^2$	by hand
Neg. Binomial (failures before r-th success, Devore)	$\binom{x+r-1}{r-1}p^r(1-p)^x$	0,1,2,...	$r(1-p)/p$	$r(1-p)/p^2$	nCr
Uniform[A, B]	$\frac{1}{B-A}$	A ≤ x ≤ B	$\frac{A+B}{2}$	$\frac{(B-A)^2}{12}$	by hand
Normal N(μ, σ²)	$\frac{1}{\sqrt{2\pi\sigma^2}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	(-∞, ∞)	μ	σ^2	Normal CD / Inverse
Exponential(λ)	$\lambda e^{-\lambda x}$	x ≥ 0	$1/\lambda$	$1/\lambda^2$	e^x key

CDFs you can write directly

Uniform: $F(x)=\frac{x-A}{B-A}$, A ≤ x ≤ B

Exponential: $F(x)=1-e^{-\lambda x}$, so $\mathrm{P}(X>x)=e^{-\lambda x}$

Normal: $F(x)=\Phi\left(\frac{x-\mu}{\sigma}\right)$ (z-table)

Binomial / Poisson: sum the pmf, or use the tables / CD

Rules for any X

$\mathrm{E}(aX+b)=a\,\mathrm{E}(X)+b$

$\mathrm{Var}(aX+b)=a^2\,\mathrm{Var}(X)$ (the +b disappears!)

$\mathrm{Var}(X)=\mathrm{E}(X^2)-[\mathrm{E}(X)]^2$

$\sigma=\sqrt{\mathrm{Var}(X)}$ (always ≥ 0)

Continuous: $\mathrm{P}(X=a)=0$ so ≤ and < are the same

C · Casio fx-991ID calculator guide

Your model has DIST mode with Normal PD/CD, Inverse Normal, Binomial PD/CD and Poisson PD/CD. Menu numbers can differ slightly between versions, so **select by name**. Each block has a **check value**: run it once tonight to confirm the input order on your unit.

1 · DIST mode: [MODE] → (scroll if needed) → DIST

Function	Gives	It asks for	Check value
Binomial PD → choose **Var**	$P(X = x)$	x, N (= n), p	x=3, N=8, p=0.4 → **0.27869**
Binomial CD → **Var**	$P(X \le x)$	x, N, p	x=3, N=8, p=0.4 → **0.59409**
Poisson PD → **Var**	$P(X = x)$	x, μ (= λ)	x=5, μ=3 → **0.10082**
Poisson CD → **Var**	$P(X \le x)$	x, μ	x=3, μ=3 → **0.64723**
Normal CD	$P(\text{Lower} \le X \le \text{Upper})$	Lower, Upper, **σ**, μ	L=485, U=515, σ=8, μ=500 → **0.93921**
Inverse Normal	x with $P(X \le x)$ =Area (**left tail only**)	Area, σ, μ	Area=0.95, σ=1, μ=0 → **1.64485**
Normal PD	the height f(x), **not a probability**	x, σ, μ	rarely needed

⚠ **TRAP Normal CD and Inverse Normal ask for σ (the SD), not σ².** The class writes X ~ N(500, 64), so enter σ = √64 = 8. In X ~ N(8,46; 0,84), σ = √0.84 = 0.9165.

▣ **CALCULATOR One-sided normal probabilities with Normal CD**
$P(X \le b)$: Lower = −1×10^99 (type [(−)] [1] [×10ˣ] [99]), Upper = b
$P(X \ge a)$: Lower = a, Upper = 1×10^99
Check: N(15015, 3540²), P(X > 18000) → **0.19955**

▣ **CALCULATOR Right-tail inverse** ("only 1% exceed")
Inverse Normal works on the left tail, so enter Area = 1 - 0.01 = 0.99.
Check: ground beef N(1, 0.10²), Area 0.99 → **1.2326** lb

▣ **CALCULATOR Turning CD output into what is asked** (discrete X): P(X = k) = PD(k) · P(X ≤ k) = CD(k) · P(X < k) = CD(k-1) · P(X ≥ k) = 1 - CD(k-1) · P(X > k) = 1 - CD(k) · P(a ≤ X ≤ b) = CD(b) - CD(a-1)

✔ Use DIST to **check** answers. On paper, still write the formula, the substitution and the z-value. The z-table method is what the class boards show (Z rounded to 2 decimals). The table answer can differ from the calculator in the 3rd decimal. Both are fine; say which you used.

2 · Integration (COMP mode): [∫▭dx] key

- [MODE] **COMP**, press [∫▭], type f(x) using [ALPHA] [)] for **X**, then the lower and upper limits with the arrow keys, then [=].
- Check: $\int_3^4 (0.075x + 0.2)\,dx$ → **0.4625**. $\int_3^5 x(0.075x + 0.2)\,dx$ → **4.05**.
- For an **infinite** limit, do the tail analytically or use a finite limit far into the tail (e.g. $\int_{0.5}^{200}$), then confirm the answer stops changing. Avoid 10^99 inside ∫, because the numeric integrator can fail.
- The class integrated the normal pdf directly with ∫. This works if the limits are within about 6σ of μ, e.g. $\int_{485}^{515} \frac{1}{\sqrt{2\pi}\cdot 64} e^{-\frac{1}{2}((x-500)/8)^2}\,dx = 0.9392$. Normal CD is faster.

3 · Solve for a percentile / median: [SHIFT] [CALC] (SOLVE)

Type the equation $F(x) = p$ with [ALPHA] [CALC] for "=". Example (median of the Practice Quiz Q6 pdf): `(X²/2+X−1.5)/6 = 0.5` → SHIFT CALC, start guess 2 → **X = 2.1623**. If two roots exist, give a starting guess inside the support [A, B].

4 · STAT mode for data: MODE → STAT → 1-VAR

- Enter the data in column x (= after each value), press AC, then SHIFT 1 (STAT) → **Var**: **n**, **x̄**, **σx** (population, ÷N), **sx** (sample, ÷(n-1)). **Use sx for a sample**; square it for s^2.
- **Sum** menu: Σx and Σx^2. These are useful to show the calculational formula on paper.
- **Frequency column** (setup: SHIFT MODE → STAT → Frequency **On**). For a pmf table, put x in column X and p(x) in FREQ. Then **x̄ = E(X)** and **σx = σ** of the random variable. Check with the dice-difference table: x̄ = 1.9444, σx^2 = 2.0525. If your unit rejects decimal frequencies, use counts instead (6, 10, 8, 6, 4, 2 out of 36). The answer is the same.
- Check for data: 2, 9, 11, 5, 6 → x̄ = 6.6, sx = 3.5071.

5 · Counting and other keys

nCr: n SHIFT ÷ r	8C3 = 56, 10C4 = 210	**nPr**: n SHIFT × r	5P3 = 60
x!: SHIFT x^{-1}	5! = 120	**e^x**: SHIFT ln	e^(-3) = 0.049787
Fraction ↔ decimal: S⇔D	70/36 ↔ 1.9444	**Ans / memory**: STO A, ALPHA A	store μ, reuse in Var

✔ Keep unrounded intermediate values in memory (STO A, B …) and round only the final answer to 4 decimals. The class boards lost accuracy by rounding early, e.g. 0.592 instead of 0.594 for P(X ≤ 3) in B(8, 0.4).

1 · Overview & descriptive statistics (Sessions 1–2, Devore Ch. 1)

Vocabulary

- **Population** *(populasi)*: all measurements of interest. Size N, mean μ, variance σ^2. **Sample** *(sampel)*: a subset. Size n, mean $\bar{x}$, variance s^2.
- **Variable** *(variabel: karakteristik / alat ukur)* vs **data** *(hasil pengukuran)*.
- **Experimental unit**: the object measured (person, light bulb).
- Uni-/bi-/multivariate: 1, 2, or more variables per unit.

Types of variables

- **Qualitative / categorical** *(non angka)*: gender, major, hair colour, ethnic group. A numeric code (1 = M, 2 = F) is still qualitative, and a student ID is qualitative too.
- **Quantitative** *(angka)*:
  - **Discrete** *(bilangan bulat)*: countable values (number of phones, semester, # of units).
  - **Continuous**: any value in an interval (body temperature 36,2°, GPA, time, weight).

Graphs

- Qualitative: **bar chart**, **pie chart**; frequency table: item | freq | relative freq (= f/n) | percent.
- Quantitative: **dotplot**, **stem-and-leaf**, **(relative-frequency) histogram**, time series (line chart).
- Histogram classes: width ≈ (max - min) / #classes, rounded **up** to a convenient number. Example: (70 - 26)/6 = 7.33 → width 8, starting at 25.

Stem-and-leaf *(batang-daun)*: (1) sort small → large *(urutkan)*; (2) split into stem | leaf, e.g. tens | ones; (3) give a key, e.g. "6 | 5 = 65" or for 1 decimal: "18 | 6 = 18.6".
Board example: 17.2, 18.6, 18.8, 18.8, 20.1, 21.3, 23.5 → `17|2 18|688 19| 20|1 21|3 22| 23|5`

Shapes *(bentuk distribusi)*

Mound-shaped, symmetric	mean ≈ median
Skewed right (long right tail, a few unusually **large** values)	**mean > median**
Skewed left (long left tail, a few unusually **small** values)	**mean < median**
Bimodal	two local peaks

Measures of centre *(ukuran pemusatan)*

Mean: $\bar{x} = \frac{\sum x_i}{n}$ (sample); $\mu = \frac{\sum x_i}{N}$ (population)

Median: the middle value after sorting. Its **position** is $0.5(n+1)$. If the position ends in .5, average the two neighbours.

Mode *(modus)*: the most frequent value. A data set can have 2 modes (bimodal) or none.

Example (slides). 2, 4, 9, 8, 6, 5, 3 → sorted 2 3 4 5 6 8 9, position 0.5(8) = 4th → median = 5.
2, 4, 9, 8, 6, 5 → sorted 2 4 5 6 8 9, position 3.5 → median = (5+6)/2 = 5.5.

Robustness (board). 2, 3, 4, 5, 6: $\bar{x}$ = 4, median 4. Add an extreme value 1000: $\bar{x}$ = 170, median = 4.5. The median is **robust** *(stabil)*, so use it for skewed data.

Measures of variability *(ukuran penyebaran)*

Range *(jangkauan)* R = max - min.

$$\sigma^2 = \frac{\sum(x_i-\mu)^2}{N} \quad \text{(population)}$$

$$s^2 = \frac{\sum(x_i-\bar{x})^2}{n-1} = \frac{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}{n-1} \quad \text{(sample)}$$

$s = \sqrt{s^2}$ is always taken positive (≥ 0). It is in the original units, while s^2 is in squared units.

We divide by n - 1 for a sample because $\bar{x}$ is estimated from the same data. This makes s^2 a better (unbiased) estimate of σ^2.

Example (slides): petals 5, 12, 6, 8, 14.
n = 5, Σx = 45, $\bar{x}$ = 9, Σx^2 = 25+144+36+64+196 = 465
$s^2 = \frac{465 - 45^2/5}{4} = \frac{465-405}{4} = 15$, $s = 3.873$.
Range = 14 - 5 = 9.

▣ **CALCULATOR** STAT 1-VAR: enter 5, 12, 6, 8, 14 → sx = 3.8730 → sx^2 = 15. Do **not** use σx (= 3.4641) for a sample.

Range approximation: R ≈ 4s to 6s, so s ≈ R/4 (a rough check).

Chebyshev (Tchebysheff) vs the Empirical rule

Chebyshev, any shape: at least $1 - \frac{1}{k^2}$ of the values lie within k SDs of the mean (k ≥ 1).
k = 2 → at least 3/4 = 75%; k = 3 → at least 8/9 ≈ 88.9%.
To find k: $k = \frac{\text{distance from mean}}{s}$.

Empirical rule, mound-shaped only: about 68% within μ ± σ, 95% within μ ± 2σ, 99.7% within μ ± 3σ.
Tails: P(above μ + 2σ) ≈ 2.5%; P(above μ + σ) ≈ 16%.

Example (slides). Task time: mean 12.8 min, SD 1.7, mound-shaped. Proportion longer than 16.2 min? 16.2 = 12.8 + 2(1.7) is 2 SDs above, so ≈ (1 - 0.95)/2 = **0.025**.

Relative standing

z-score: $z = \frac{x - \bar{x}}{s}$. |z| ≤ 2 is not unusual; |z| > 3 suggests an outlier. **p-th percentile**: p% of the values lie below it.

Quartiles, IQR, five-number summary, box plot

After sorting: position of $Q_1 = 0.25(n+1)$, median $= 0.5(n+1)$, $Q_3 = 0.75(n+1)$. If a position is fractional, **interpolate**: value at 4.75 = $x_{(4)}$ + 0.75($x_{(5)}$ - $x_{(4)}$).
$IQR = Q_3 - Q_1$
Fences *(batas)*: LL = $Q_1 - 1.5\,IQR$, UL = $Q_3 + 1.5\,IQR$. Points beyond them are **outliers**, marked *.
Five-number summary: Min, Q1, Median (Q2), Q3, Max.

Box plot: box from Q1 to Q3 with a line at the median. Whiskers go to the most extreme values that are *not* outliers.
Median left of centre and a long right whisker means skewed right. Median right of centre and a long left whisker means skewed left.

Example (slides): shoe prices, n = 18
40 60 65 65 65 68 68 70 70 70 70 70 70 74 75 75 90 95
Q1 position = 0.25(19) = 4.75 → 65 + 0.75(65 - 65) = **65**
Q3 position = 0.75(19) = 14.25 → $x_{(14)}$ + 0.25($x_{(15)}$ - $x_{(14)}$) = 74 + 0.25(75 - 74) = **74.25**
Median at 9.5 → 70. IQR = 9.25; fences 51.125 and 88.125 → **40, 90 and 95 are outliers**.
(The slide gets 75.25 because its sorted list leaves out the 40. Use your own sorted list and count positions carefully.)

Example (slides): sodium in 8 cheeses
260 290 300 320 330 340 340 520
Q1 position 2.25 → 290 + 0.25(10) = 292.5; Q3 position 6.75 → 340 + 0.75(0) = 340
IQR = 47.5; LL = 292.5 - 71.25 = 221.25; UL = 340 + 71.25 = 411.25
→ **520 is an outlier**. The upper whisker stops at 340.

⚠ **TRAP** Different textbooks and calculators use different quartile rules. In the exam, use the **0.25(n+1)** rule from the slides and show the position you computed.

2 · Probability (Sessions 3–4, Devore Ch. 2)

Sample spaces and events

- **Experiment** *(percobaan)*: a process that produces an outcome. **Sample space S** *(ruang sampel)*: the set of all simple outcomes.
- **Event** *(kejadian)*: a subset of S, written with a capital letter.
- $A \cup B$: A or B or both *(gabungan)*. $A \cap B$: both *(irisan)*. A^c (also A′): not A *(komplemen)*.
- **Mutually exclusive** *(saling lepas)*: $A \cap B = \emptyset$, so P(A ∩ B) = 0.
- De Morgan: $(A \cup B)^c = A^c \cap B^c$, $(A \cap B)^c = A^c \cup B^c$.

Axioms and basic rules

1. $\mathrm{P}(A) \geq 0$ · 2. $\mathrm{P}(S) = 1$ · 3. If $A_1, A_2, \ldots$ are mutually exclusive: $\mathrm{P}(\cup A_i) = \sum \mathrm{P}(A_i)$
→ $0 \leq \mathrm{P}(A) \leq 1$, $\mathrm{P}(\emptyset) = 0$, $\mathrm{P}(A^c) = 1 - \mathrm{P}(A)$
Equally likely outcomes: $\mathrm{P}(A) = \dfrac{n(A)}{n(S)}$

Relative-frequency interpretation: P(A) is the long-run proportion n(A)/n as n → ∞.

Board: n(S) = 100 people, n(A) = 25 wear glasses → P(A) = 25/100 = 0.25.

Addition rule, complement, "at least one"

$$\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B) - \mathrm{P}(A \cap B)$$

Mutually exclusive: $\mathrm{P}(A \cup B) = \mathrm{P}(A) + \mathrm{P}(B)$
Three events: $\mathrm{P}(A \cup B \cup C) = \mathrm{P}(A) + \mathrm{P}(B) + \mathrm{P}(C) - \mathrm{P}(AB) - \mathrm{P}(AC) - \mathrm{P}(BC) + \mathrm{P}(ABC)$

"At least one" ***(sedikitnya satu)***: $\mathrm{P}(\geq 1) = 1 - \mathrm{P}(\text{none})$
If the trials are independent with success prob p each: $1 - (1-p)^n$.
Example: 3 independent alarms each detect with 0.85: $1 - 0.15^3 = 0.9966$.

Example (slides): 120 students. A = Javanese (50), B = female (60), both = 30. $\mathrm{P}(A \cup B) = \frac{50+60-30}{120} = \frac{80}{120} = \frac{2}{3}$.

Counting techniques

Situation	Formula	Example
mn rule (product rule): k stages with $n_1, n_2, \ldots$ ways	$n_1 n_2 \cdots n_k$	Toss 2 dice: 6 × 6 = 36. Three coins: $2^3 = 8$.
Permutation: choose r of n, **order matters** (arrange, rank, code, president/secretary)	$P_r^n = \dfrac{n!}{(n-r)!}$	3-digit lock code from {1,2,3,4}, no repeats: 4P3 = 24. Arrange 5 parts: 5! = 120.
Combination: choose r of n, **order does not matter** (committee, hand, sample)	$C_r^n = \binom{n}{r} = \dfrac{n!}{r!(n-r)!}$	3 of 5 people for a subcommittee: 5C3 = 10.
With repetition allowed, order matters	n^r	4-digit PIN: 10^4
Arrange n objects with identical groups $n_1, n_2, \ldots$	$\dfrac{n!}{n_1!\, n_2! \cdots}$	"STATISTIK": 9!/(2!·3!·2!) for S, T, I

Example (slides): box of 6 M&Ms, 4 red and 2 green, choose 2. P(exactly one red) = $\dfrac{\binom{4}{1}\binom{2}{1}}{\binom{6}{2}} = \dfrac{8}{15} = 0.5333$.

Order does not matter, so use combinations in both the numerator and the denominator.

✔ To decide between a permutation and a combination, ask: "If I swap two chosen items, is it a different result?" If yes, it is a permutation. If no, it is a combination.

Contingency tables: joint, marginal, conditional *(tabel kontingensi)*

This was the most-practised type on the class boards. Always write the table **with totals** first.

Age \ Health insurance	Yes	No	Total
18–34	750	170	920
35 & older	950	130	1080
Total	1700	300	2000

Joint (cell ÷ grand total): $\mathrm{P}(18\text{–}34 \cap \mathrm{Yes}) = 750/2000 = 0.375$
Marginal (row or column total ÷ grand total): $\mathrm{P}(\mathrm{No}) = 300/2000 = 0.15$, $\mathrm{P}(18\text{–}34) = 920/2000 = 0.46$
Conditional (cell ÷ total of the *given* row or column):
$\mathrm{P}(\mathrm{No} \mid 18\text{–}34) = 170/920 = 0.1848$
$\mathrm{P}(\mathrm{No} \mid 35{+}) = 130/1080 = 0.1204$
$\mathrm{P}(18\text{–}34 \mid \mathrm{No}) = 170/300 = 0.5667$

Reading P(··· | ···): the left side is what is asked *(ditanya / dicari)*. The right side is what is known, the condition *(diketahui / syarat)*.
The denominator is always the total of the **condition**.

Sports example (class, Anderson p.191, millions)
Walking: M 28.7, F 57.7, total 86.4. All: M 123.3, F 161.8, total 285.1.
P(walking) = 86.4/285.1 = 0.3031
P(walking | female) = 57.7/161.8 = 0.3566
P(woman | walker) = 57.7/86.4 = 0.6678. This is **conditional**, not joint.
P(woman ∩ walker) = 57.7/285.1 = 0.2024

✔ Independence test from a table: compare $\mathrm{P}(\mathrm{No} \mid 18\text{–}34) = 0.1848$ with $\mathrm{P}(\mathrm{No}) = 0.15$. They differ, so age and insurance are **not independent**.

Conditional probability *(peluang bersyarat)*

$$\mathrm{P}(A \mid B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)}, \qquad \mathrm{P}(B) > 0$$

$\mathrm{P}(A \mid B) \neq \mathrm{P}(B \mid A)$ in general, but $\mathrm{P}(A \cap B) = \mathrm{P}(B \cap A)$.
$\mathrm{P}(A^c \mid B) = 1 - \mathrm{P}(A \mid B)$

Example (slides): 5 M&Ms, 2 red and 3 blue, draw 2 without replacement.
P(2nd red | 1st blue) = 2/4 = 1/2; P(2nd red | 1st red) = 1/4. The answer depends on the 1st draw, so the events are **dependent**.

Independence *(saling bebas)* vs mutually exclusive *(saling lepas)*

A and B are **independent** if any one of these holds (then all do):
1. $\mathrm{P}(A \mid B) = \mathrm{P}(A)$ · 2. $\mathrm{P}(B \mid A) = \mathrm{P}(B)$ · 3. $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$
If A and B are independent, so are A^c and B, A and B^c, and A^c and B^c.

⚠ **TRAP** Mutually exclusive events with P(A), P(B) > 0 are **never** independent: P(A ∩ B) = 0 ≠ P(A)P(B).
"Saling lepas" is about overlap. "Saling bebas" is about one event giving no information about the other.

Multiplication rule

$\mathrm{P}(A \cap B) = \mathrm{P}(A)\ \mathrm{P}(B \mid A) = \mathrm{P}(B)\ \mathrm{P}(A \mid B)$
Independent: $\mathrm{P}(A \cap B) = \mathrm{P}(A)\,\mathrm{P}(B)$
Chain: $\mathrm{P}(A \cap B \cap C) = \mathrm{P}(A)\,\mathrm{P}(B \mid A)\,\mathrm{P}(C \mid A \cap B)$

Example (slides): 10% high-risk. 3 people: P(exactly one high-risk) = HNN + NHN + NNH = $3(0.1)(0.9)^2$ = **0.243**. This is the Binomial B(3, 0.1) at x = 1.
49% female, P(H | F) = 0.08 → P(H ∩ F) = 0.49 × 0.08 = **0.0392**.

Law of total probability

If $B_1, \ldots, B_k$ are mutually exclusive and exhaustive (a partition: exactly one happens):

$$\mathrm{P}(A) = \sum_{i=1}^{k} \mathrm{P}(B_i)\ \mathrm{P}(A \mid B_i)$$

Draw a **tree**: the first branches are B_i with $\mathrm{P}(B_i)$, the second branches are A | B_i. Multiply along each path, then add the paths that end in A.

Bayes' rule

$$\mathrm{P}(B_j \mid A) = \frac{\mathrm{P}(B_j)\ \mathrm{P}(A \mid B_j)}{\sum_i \mathrm{P}(B_i)\ \mathrm{P}(A \mid B_i)} = \frac{\text{the one path}}{\text{all paths to } A}$$

Prior $\mathrm{P}(B_j)$ → posterior $\mathrm{P}(B_j \mid A)$ after observing A.

Example (slides): heart attack risk. P(F) = 0.49, P(M) = 0.51, P(H | F) = 0.08, P(H | M) = 0.12. A randomly selected person is high-risk. P(male)?
P(H) = 0.49(0.08) + 0.51(0.12) = 0.0392 + 0.0612 = 0.1004
$\mathrm{P}(M \mid H) = \frac{0.0612}{0.1004} = \mathbf{0.6096}$; $\mathrm{P}(F \mid H) = 0.3904$. Check: the two add to 1.

Template: machines. Machines A, B, C make 50%, 30%, 20% of the output, with defect rates 2%, 3%, 5%.
P(D) = 0.5(0.02) + 0.3(0.03) + 0.2(0.05) = 0.010 + 0.009 + 0.010 = 0.029
P(C | D) = 0.010/0.029 = 0.3448 · P(B | D) = 0.3103 · P(A | D) = 0.3448

✔ Bayes in a table: build a table of joint probabilities, $\mathrm{P}(B_i)\,\mathrm{P}(A \mid B_i)$ for A and for A^c. Then every posterior is "cell ÷ column total", exactly like the contingency tables above.

3 · Discrete random variables (Sessions 5–8, Devore Ch. 3)

Random variable *(variabel random)*

A **function** that maps each outcome of an experiment to a number *(fungsi yang memetakan outcome ke bilangan)*. It is written with a capital letter (X), and its values with a lower-case letter (x).

- Pass/fail status: X = {lulus, tidak lulus} → coded {1, 0}. This is a **Bernoulli** random variable.
- Number of phones owned: X = {0, 1, 2, ...} → **discrete**.
- Midterm score 0 ≤ Z ≤ 100 (any real number) → **continuous** (Ch. 4).

pmf and cdf

pmf *(fungsi peluang / probability mass function)*: $p(x) = \mathrm{P}(X = x)$
Valid if: $0 \le p(x) \le 1$ and $\sum_x p(x) = 1$.
cdf *(fungsi kumulatif)*: $F(x) = \mathrm{P}(X \le x) = \sum_{y \le x} p(y)$. It is a step function, non-decreasing, rising from 0 to 1.
$\mathrm{P}(a \le X \le b) = F(b) - F(a^-)$, where a^- is the possible value just below a.
$p(x) = F(x) - F(x^-)$, the size of the jump at x.

Coin tossed 3 times, X = # heads (class)

8 equally likely outcomes, each 1/8: HHH(3) HHT(2) HTH(2) THH(2) HTT(1) THT(1) TTH(1) TTT(0)

x	0	1	2	3
p(x)	1/8	3/8	3/8	1/8
F(x)	1/8	4/8	7/8	1

$$F(x) = \begin{cases} 0 & x < 0 \\ 1/8 & 0 \le x < 1 \\ 4/8 & 1 \le x < 2 \\ 7/8 & 2 \le x < 3 \\ 1 & x \ge 3 \end{cases}$$

Table with an unknown constant

p(x): x = 0, 1, 2, 3 → 0.1, k, 2k, 0.3. Use Σ = 1: 0.4 + 3k = 1, so k = 0.2.
Always check that every p(x) is between 0 and 1 afterwards.

Formula pmf: p(x) = c·x for x = 1, 2, 3, 4 → c(1+2+3+4) = 1 → c = 1/10.

Expected value and variance

$$\mu = \mathrm{E}(X) = \sum_x x\,p(x) \qquad \mathrm{E}[h(X)] = \sum_x h(x)\,p(x)$$

$$\sigma^2 = \mathrm{Var}(X) = \sum (x-\mu)^2 p(x) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2$$

$\sigma = \sqrt{\sigma^2}$
$\mathrm{E}(aX+b) = a\mu + b$, $\mathrm{Var}(aX+b) = a^2\sigma^2$, $\sigma_{aX+b} = |a|\sigma$

Class contoh soal: X = positive difference of 2 dice *(selisih positif 2 mata dadu)*

Count the 36 pairs: X=0: 6, X=1: 10, X=2: 8, X=3: 6, X=4: 4, X=5: 2.

x	0	1	2	3	4	5	Σ
p(x)	6/36	10/36	8/36	6/36	4/36	2/36	1
x·p	0	10/36	16/36	18/36	16/36	10/36	70/36
x²·p	0	10/36	32/36	54/36	64/36	50/36	210/36

μ = 70/36 = **1.9444**; E(X²) = 210/36 = 5.8333; σ² = 5.8333 - 1.9444² = **2.0525**; σ = 1.4327

Fair die (slides): μ = 21/6 = 3.5, E(X²) = 91/6, σ² = 91/6 - 12.25 = 35/12 = 2.9167, σ = 1.7078.

Binomial distribution *(distribusi binomial)*: X ~ B(n, p)

Check all four conditions (class board):
1. each trial has only **2 outcomes** (success/failure)
2. a **fixed number** of trials n
3. trials are **independent**
4. P(success) = p is **known and constant** on every trial

X = number of successes in n trials.

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \ldots, n$$

$\mu = np$, $\sigma^2 = np(1-p)$ (q = 1 - p)

⚠ **TRAP** Sampling **without replacement** from a small population breaks independence, so use the Hypergeometric. If the sample is ≤ 5% of the population, the Binomial is OK (Devore).

▣ **CALCULATOR** Binomial PD (x, N, p) for P(X = x). Binomial CD for P(X ≤ x). For P(X ≥ k), use 1 - CD(k-1).

Marksman (slides): hits 80%, 5 shots. P(X = 3) = $10(0.8)^3(0.2)^2$ = 0.2048. P(X > 3) = P(4) + P(5) = 0.4096 + 0.3277 = 0.7373. μ = 4, σ = √0.8 = 0.894. Zero hits is (0 - 4)/0.894 = -4.5 SDs from the mean, so it would be very unusual.

Contoh soal: P(pass) = 0,4, 8 students. X ~ B(8, 0.4). class board

1. P(exactly 3 pass) = $\binom{8}{3}(0.4)^3(0.6)^5 = 56(0.064)(0.07776) =$ **0.2787**

2. P(2–4 pass) = P(2) + P(3) + P(4) = 0.2090 + 0.2787 + 0.2322 = **0.7200**

3. P(at most 3) ***(paling banyak 3)*** = P(0) + P(1) + P(2) + P(3) = 0.0168 + 0.0896 + 0.2090 + 0.2787 = **0.5941** (the board's 0.592 comes from rounding too early)

4. P(at least 8) ***(paling sedikit 8)*** = P(X = 8) = $(0.4)^8$ = **0.00066**

Mean = 8(0.4) = 3.2, variance = 8(0.4)(0.6) = 1.92.

x	0	1	2	3	4	5	6	7	8
p(x)	.0168	.0896	.2090	.2787	.2322	.1239	.0413	.0079	.0007

X ~ B(10, 0.25): a) P(X = 5) = $252(0.25)^5(0.75)^5$ = **0.0584** · b) P(X ≥ 4) = 1 - [P(0) + P(1) + P(2) + P(3)] = 1 - (0.0563 + 0.1877 + 0.2816 + 0.2503) = 1 - 0.7759 = **0.2241** · c) P(X = 0) = 0.75^{10} = **0.0563** · d) P(X ≤ 2) = **0.5256** class board

Poisson distribution *(distribusi Poisson)*: X ~ Poisson(λ)

X = **number of events** in a fixed interval of time or space (calls per hour, typos per page, accidents per week, vehicles through a toll gate).

λ = average number of events **in that interval** *(rata-rata kejadian)*.

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!}, \quad x = 0, 1, 2, \ldots$$

$\mu = \lambda$, $\sigma^2 = \lambda$

Rescale λ: rate α per unit, interval length t → λ = αt. Approximates the Binomial when n is large and p is small: λ = np (Devore: n > 50, np < 5).

Typos: λ = 3 per page. class board

1. P(X = 5) = $e^{-3}3^5/5!$ = **0.1008**
2. P(X ≤ 3) = $e^{-3}(1 + 3 + 9/2 + 27/6)$ = 0.0498 + 0.1494 + 0.2240 + 0.2240 = **0.6472**
3. P(X ≥ 2) = 1 - P(0) - P(1) = 1 - 0.0498 - 0.1494 = **0.8009**

Accidents (slides): λ = 2 per week. P(X = 1) = $2e^{-2}$ = 0.2707. P(X ≥ 8) = 1 - P(X ≤ 7) = 1 - 0.9989 = 0.0011. For **2 weeks**, λ = 4.

Hypergeometric distribution: sampling without replacement *(tanpa pengembalian)*

Population of N items, of which M are "successes". Draw n without replacement. X = number of successes in the sample.

$$\mathrm{P}(X = x) = \frac{\binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}}$$

$$\mathrm{E}(X) = n\frac{M}{N},\ \mathrm{Var}(X) = \frac{N-n}{N-1} \cdot n\frac{M}{N}\left(1 - \frac{M}{N}\right)$$

The factor (N - n)/(N - 1) is the finite-population correction. Without it, the variance is the Binomial one with p = M/N.

Batteries (slides). A pack of 8 has 2 defective. Pick 4. Success = a working battery: N = 8, M = 6, n = 4.

P(all 4 work) = $\frac{\binom{6}{4}\binom{2}{0}}{\binom{8}{4}} = \frac{15}{70} = \mathbf{0.2143}$

E(X) = 4(6/8) = 3; Var(X) = (4/7)(4)(0.75)(0.25) = 0.4286.

Geometric and Negative Binomial (listed in the sub-topics; low priority)

Geometric: X = trial number of the **first** success. $\mathrm{P}(X = x) = (1-p)^{x-1}p$, x = 1, 2, ...
$\mathrm{P}(X > k) = (1-p)^k$, $\mathrm{E}(X) = 1/p$, $\mathrm{Var}(X) = (1-p)/p^2$.
Example: p = 0.1. P(first success on the 3rd try) = $0.9^2(0.1)$ = 0.081. P(success within 3 tries) = $1 - 0.9^3$ = 0.271.

Negative Binomial (Devore's form): X = number of **failures** before the r-th success.
$\mathrm{P}(X = x) = \binom{x+r-1}{r-1}p^r(1-p)^x$, $\mathrm{E}(X) = \frac{r(1-p)}{p}$, $\mathrm{Var}(X) = \frac{r(1-p)}{p^2}$.
Example: r = 2, p = 0.2, x = 3 failures: $\binom{4}{1}(0.2)^2(0.8)^3 = 0.0819$.
Other books count total trials X + r. Check which definition the question uses.

Discrete checklist

1. Define X in words *(X = banyaknya ...)* and list its possible values. The class boards always do this.
2. Name the distribution and its parameters: "X ~ B(8, 0.4)", "X ~ Poisson(3)".
3. Translate the words into an inequality (Tab G), and turn it into = / ≤ / complement form.
4. Write the formula, substitute, and compute. Check with DIST.
5. Sanity check: is 0 ≤ P ≤ 1? Is it near the mean, which is where the largest probabilities are?

4 · Continuous random variables (Sessions 9–12, Devore Ch. 4)

X takes any real value in an interval *(bilangan real, dinyatakan dalam interval / range)*: midterm score 0 ≤ x ≤ 100, height 140 ≤ y ≤ 200, room temperature 20 ≤ z ≤ 38.

Probability density function (pdf)

$$\mathrm{P}(a \le X \le b) = \int_a^b f(x)\,dx \quad (\text{area under } f)$$

f is a valid pdf iff *(syarat pdf)*:
(1) $f(x) \ge 0$ for all x, and (2) $\int_{-\infty}^{\infty} f(x)\,dx = 1$.
Consequences: $\mathrm{P}(X = a) = 0$, so
$\mathrm{P}(a \le X \le b) = \mathrm{P}(a < X < b) = \mathrm{P}(a \le X < b) = \dots$
f(x) itself is **not** a probability and can be > 1.

"Buktikan … pdf" / "find c": template

1. State that f(x) ≥ 0 on the support (e.g. "e^(·) > 0 and 0.15 > 0", or check the endpoints of a linear f).
2. Integrate over the support only (f = 0 elsewhere): $\int_{-\infty}^{\infty} f = \int_{\text{support}} f + \int_{\text{elsewhere}} 0$.
3. Show the result = 1 (or solve c from "= 1").
4. Conclude: "∴ f(x) merupakan pdf" (∴ f(x) is a pdf).

⚠ **TRAP** If ∫ f ≠ 1, the function is **not** a pdf. Do not "fix" it by dividing by the total area unless the question says f is proportional to … (∝). See the class example on **p.19**.

Contoh soal 1: prove that $f(x) = 0.15e^{-0.15(x-0.5)}$ for x ≥ 0.5 (0 otherwise) is a pdf. class board / Devore Ex. 4.5
(time headway between cars)
(1) For x ≥ 0.5, 0.15 > 0 and the exponential is > 0, so f(x) ≥ 0. Elsewhere f = 0 ≥ 0.
(2) $\int_{-\infty}^{\infty} f(x)dx = \int_{0.5}^{\infty} 0.15e^{-0.15(x-0.5)}dx = \left[-e^{-0.15(x-0.5)}\right]_{0.5}^{\infty} = 0 - (-e^0) = 1$ ✓ ∴ f is a pdf.
Extra: P(X ≤ 5) = $\left[-e^{-0.15(x-0.5)}\right]_{0.5}^{5} = 1 - e^{-0.675} = \mathbf{0.4908}$. cdf: $F(x) = 1 - e^{-0.15(x-0.5)}$ for x ≥ 0.5. This is an exponential distribution with λ = 0.15 shifted right by 0.5, so E(X) = 0.5 + 1/0.15 = 7.1667 and Var(X) = $1/0.15^2$ = 44.44.

Cumulative distribution function (cdf) *(fungsi kumulatif)*

$$F(x) = \mathrm{P}(X \le x) = \int_{-\infty}^{x} f(t)\,dt$$

$\mathrm{P}(a \le X \le b) = F(b) - F(a) \cdot \mathrm{P}(X > a) = 1 - F(a)$
$f(x) = F'(x)$ wherever the derivative exists
F is continuous and non-decreasing, with F(-∞) = 0 and F(∞) = 1.

Writing a piecewise cdf (support [A, B]): $F(x) = \begin{cases} 0 & x < A \\ \int_A^x f(t)\,dt & A \le x \le B \\ 1 & x > B \end{cases}$
Check: the middle piece must be 0 at x = A and 1 at x = B.

Gravel sales (tons/week): $f(x) = 1.5(1 - x^2)$, 0 ≤ x ≤ 1. slides / Devore Ex. 4.7
$F(x) = \int_0^x 1.5(1 - t^2)dt = 1.5\left(x - \frac{x^3}{3}\right)$ for 0 ≤ x ≤ 1 (0 below, 1 above).
P(X ≤ 0.25) = 1.5(0.25 - 0.005208) = **0.3672**; P(X > 0.75) = 1 - 1.5(0.75 - 0.140625) = **0.0859**; P(0.25 < X < 0.75) = 0.9141 - 0.3672 = **0.5469**
E(X) = $\int_0^1 x \cdot 1.5(1 - x^2)dx = 1.5\left(\frac{1}{2} - \frac{1}{4}\right) = \mathbf{0.375}$; E(X²) = 1.5(1/3 - 1/5) = 0.2; Var = 0.2 - 0.375^2 = **0.0594**; σ = 0.2437

Percentiles and the median

The (100p)-th percentile η(p) solves $F(\eta(p)) = p$.
Median $\tilde{\mu}$: $F(\tilde{\mu}) = 0.5$. Quartiles: F = 0.25 and F = 0.75.
For a symmetric pdf, median = mean = centre of symmetry.

Gravel, 40th percentile: 1.5(η - η^3/3) = 0.4 → η^3 - 3η + 0.8 = 0 → **η = 0.2735** (SHIFT SOLVE, start at 0.3). Median: 1.5(m - m^3/3) = 0.5 → m = 0.3473.

Expected value and variance (continuous)

$$\mu = \mathrm{E}(X) = \int_{-\infty}^{\infty} x\, f(x)\, dx$$

$$\mathrm{E}[h(X)] = \int h(x) f(x)\, dx$$

$$\sigma^2 = \mathrm{Var}(X) = \int (x-\mu)^2 f(x)\, dx = \mathrm{E}(X^2) - \mu^2$$

$\mathrm{E}(X^2) = \int x^2 f(x)\, dx$ · the rules for aX + b are the same as in the discrete case.

✔ Each integral runs over the **support only**. Remember the extra x (or x^2) inside. The most common slip is computing ∫ f(x) dx (= 1) when you meant ∫ x·f(x) dx.

Uniform distribution *(distribusi seragam)*: X ~ U[A, B]

$f(x) = \frac{1}{B-A}$ for A ≤ x ≤ B · $F(x) = \frac{x-A}{B-A}$

$\mathrm{P}(c \le X \le d) = \frac{d-c}{B-A}$ (length ratio)

$\mathrm{E}(X) = \frac{A+B}{2}$, $\mathrm{Var}(X) = \frac{(B-A)^2}{12}$, median = (A + B)/2

Tire imperfection angle (slides): X ~ U[0, 360). P(within 90° of the reference line) = P(0 ≤ X ≤ 90) + P(270 ≤ X < 360) = 90/360 + 90/360 = **0.5**.
Bus every 20 min: wait W ~ U[0, 20]. P(W > 15) = 5/20 = 0.25. E = 10 min, Var = 400/12 = 33.33.

Given f(x) = 0,075x + 0,2 for 3 ≤ x ≤ 5 (0 otherwise). Find a) P(X ≤ 4) b) P(X > 4) c) P(3.5 ≤ X ≤ 4.5) d) P(X > 4.5) e) μ f) σ². class board, contoh soal 2

Note: the board reads "0,75x", but that integrates to 6.4 over [3, 5], so it is not a valid pdf. With **0,075** the integral is exactly 1, which is clearly what was intended. In class the students divided by 6.4 instead. That gives different numbers, e.g. 0.441 vs 0.4625 for (a).

Validity: f(3) = 0.425 > 0 and f(5) = 0.575 > 0, and f is linear, so f ≥ 0 on [3, 5]. $\int_3^5 (0.075x + 0.2)dx = \left[0.0375x^2 + 0.2x\right]_3^5 = (0.9375 + 1) - (0.3375 + 0.6) = 1$ ✓

Antiderivative G(x) = 0.0375x² + 0.2x. **cdf**: F(x) = 0.0375x² + 0.2x - 0.9375 for 3 ≤ x ≤ 5.

a) P(X ≤ 4)	$\int_3^4 f = G(4) - G(3) = 1.4 - 0.9375$	**0.4625**
b) P(X > 4)	$\int_4^5 f + \int_5^\infty 0 = 1.9375 - 1.4$ (or 1 - 0.4625)	**0.5375**
c) P(3.5 ≤ X ≤ 4.5)	$G(4.5) - G(3.5) = 1.659375 - 1.159375$	**0.5000**
d) P(X > 4.5)	$G(5) - G(4.5) = 1.9375 - 1.659375$	**0.2781**
e) μ	$\int_3^5 x(0.075x + 0.2)dx = \left[0.025x^3 + 0.1x^2\right]_3^5 = (3.125 + 2.5) - (0.675 + 0.9)$	**4.05**
E(X²)	$\int_3^5 x^2(0.075x + 0.2)dx = \left[0.01875x^4 + \frac{0.2}{3}x^3\right]_3^5 = (11.71875 + 8.3333) - (1.51875 + 1.8)$	16.7333
f) σ²	$16.7333 - 4.05^2 = 16.7333 - 16.4025$	**0.3308**
(extra) median	$F(m) = 0.5 \Rightarrow 0.0375m^2 + 0.2m - 1.4375 = 0 \Rightarrow m = \frac{-0.2+\sqrt{0.04+0.215625}}{0.075}$	4.0746

Sanity check: the density increases with x, so the mean (4.05) is a little right of the midpoint 4, and P(X > 4) > 0.5. ✓

Discrete ↔ continuous dictionary

	Discrete	Continuous
probability function	pmf p(x) = P(X = x)	pdf f(x); P(X = x) = 0
total	Σ p(x) = 1	∫ f(x) dx = 1
cdf	$F(x) = \Sigma_{y \le x}\, p(y)$ (steps)	$F(x) = \int_{-\infty}^{x} f$ (smooth)
mean / E(X²)	Σ x p(x) / Σ x² p(x)	∫ x f(x) dx / ∫ x² f(x) dx
≤ vs <	**matters**	does not matter

5 · Normal distribution *(distribusi normal)*

$X \sim N(\mu, \sigma^2)$ with parameters μ (mean, -∞ < μ < ∞) and **σ² (variance) > 0**. Domain (-∞, ∞).

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$\mathrm{P}(X \le x) = \int_{-\infty}^{x} f(t)\,dt$. This has no closed form, so we standardize and use the z-table (or Normal CD). The curve is bell-shaped and symmetric about μ: mean = median = mode. A larger σ makes it wider and flatter.

⚠ **TRAP N(500, 64) means σ² = 64, so σ = 8.** The class writes the variance as the second parameter. Always take the square root before standardizing. If the question says "simpangan baku 8", then σ = 8 directly.

Empirical rule: P(|Z| ≤ 1) = 0.6826, P(|Z| ≤ 2) = 0.9544, P(|Z| ≤ 3) = 0.9974.
Symmetry: Φ(-z) = 1 - Φ(z); P(Z > z) = Φ(-z).

Standard normal Z ~ N(0, 1) *(normal baku)* and how to use the table

$$Z = \frac{X-\mu}{\sigma} \sim N(0,1), \qquad f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Steps (class board):
1. Convert x to z: $z = (x-\mu)/\sigma$
2. Round z to **2 decimals** *(Z hanya boleh 2 desimal)*
3. Read Φ(z) = area to the **left** from Tab T (**p.24**).

Asked	Use
$\mathrm{P}(X < x) = \mathrm{P}(X \le x)$	$\Phi(z)$ directly *(langsung lihat tabel)*
$\mathrm{P}(X > x) = \mathrm{P}(X \ge x)$	$1 - \Phi(z)$
$\mathrm{P}(a < X < b)$	$\Phi(z_b) - \Phi(z_a)$
$\mathrm{P}(X = x)$	**0** (continuous)
z > 3.49 (off the table)	Φ(z) ≈ 1
z < -3.49	Φ(z) ≈ 0

Semiconductor thickness: mean 500 nm, variance 64 → X ~ N(500, 64), σ = 8. Spec is 485–515 nm. class board
1. $\mathrm{P}(485 \le X \le 515)$: $z = \frac{485-500}{8} = -1.875 \to -1.88$, $z = \frac{515-500}{8} = 1.875 \to 1.88$
 = Φ(1.88) - Φ(-1.88) = 0.9699 - 0.0301 = **0.9398** (Normal CD / ∫ gives 0.9392)
2. $\mathrm{P}(X \le 510)$: z = 10/8 = 1.25 → Φ(1.25) = **0.8944**

X ~ N(15015, 3540²), so σ = 3540. class problem "17"
a) P(X > 18000): z = 2985/3540 = 0.84 → 1 - 0.7995 = **0.2005**
b) P(X < 10000): z = -5015/3540 = -1.42 → **0.0778**
c) P(12000 < X < 18000): z = -0.85 and 0.84 → 0.7995 - 0.1977 = **0.6018** (exact 0.6033)
d) P(X ≤ 14000): z = -1015/3540 = -0.29 → **0.3859** (exact 0.3872)

X ~ N(8.46, 0.84): σ² = 0.84, so σ = √0.84 = 0.9165. class board
P(8 ≤ X ≤ 10): z = (10 - 8.46)/0.9165 = 1.68, z = (8 - 8.46)/0.9165 = -0.50 → 0.9535 - 0.3085 = **0.6450**
P(X ≥ 10) = 1 - 0.9535 = **0.0465** · P(X > 15): z = 7.14 > 3.49 → ≈ **0**
(The board used σ = 0.913, which gives z = 1.69, 0.6460 and 0.0455. Use √0.84 = 0.9165.)

Slides: X ~ N(5, 2²): P(X > 7) = 1 - Φ(1) = 0.1587. Ground beef N(1, 0.10²): P(0.80 < X < 0.85) = Φ(-1.5) - Φ(-2) = 0.0668 - 0.0228 = **0.0440**.

Working backwards: percentiles *(mencari nilai x dari peluang)*

1. Turn the statement into a **left** area: "top 1%" or "only 1% exceed" → left area 0.99.
2. Find z in the **body** of the table with area closest to it. If it is halfway between two entries, average the z values.
3. $x = \mu + z\sigma$. In general $\eta(p) = \mu + z_p\,\sigma$.

Left area	z	Left area	z
0.01	-2.33	0.90	1.28
0.025	-1.96	0.95	1.645
0.05	-1.645	0.975	1.96
0.10	-1.28	0.99	2.33
0.25 (Q1)	-0.67	0.995	2.58
0.50	0	0.75 (Q3)	0.67

Slides: the z with 0.05 to its right: left area 0.95 is halfway between 0.9495 (1.64) and 0.9505 (1.65), so z = **1.645**.
Ground beef, a weight that only 1% exceed: z = 2.33 → x = 1 + 2.33(0.10) = **1.233 lb** (Inverse Normal: 1.2326).
Semiconductor: thickness of the thinnest 10%: x = 500 - 1.28(8) = **489.76 nm**.

Normal approximation to the Binomial

If X ~ B(n, p) with **np ≥ 5 and n(1 - p) ≥ 5** (slides: "both greater than 5"; Devore uses 10), then X ≈ N(μ = np, σ^2 = np(1 - p)).
Continuity correction: each integer k covers [k - 0.5, k + 0.5].

Binomial	Normal (with correction)
P(X ≤ k)	P(Y ≤ k + 0.5)
P(X < k)	P(Y ≤ k - 0.5)
P(X ≥ k)	P(Y ≥ k - 0.5)
P(X > k)	P(Y ≥ k + 0.5)
P(X = k)	P(k - 0.5 ≤ Y ≤ k + 0.5)
P(a ≤ X ≤ b)	P(a - 0.5 ≤ Y ≤ b + 0.5)

Slides: n = 30, p = 0.4: np = 12, nq = 18, so the approximation is OK. σ = √7.2 = 2.683. P(X ≤ 10) ≈ P(Y ≤ 10.5) = Φ((10.5 - 12)/2.683) = Φ(-0.56) = **0.2877** (exact Binomial 0.2915).
Batteries: n = 200, p = 0.95: np = 190, nq = 10, so OK. σ = √9.5 = 3.082. P(X ≥ 195) ≈ P(Y ≥ 194.5) = 1 - Φ(1.46) = 1 - 0.9279 = **0.0721** (exact 0.0623; nq is small, so the approximation is rough).

✔ You have Binomial CD in DIST mode. If the question explicitly says "gunakan pendekatan normal" (use the normal approximation), do the approximation. Otherwise give the exact Binomial and mention the approximation as a check.

6 · Exponential distribution *(distribusi eksponensial)*

$X \sim \text{Exp}(\lambda)$, where λ > 0 is the **rate** *(laju)*. Domain x ≥ 0.

$$f(x) = \lambda e^{-\lambda x},\ x \geq 0 \qquad F(x) = 1 - e^{-\lambda x}$$

$\text{P}(X > x) = e^{-\lambda x}$ · $\text{P}(a \leq X \leq b) = e^{-\lambda a} - e^{-\lambda b}$

$\text{E}(X) = \dfrac{1}{\lambda}$, $\text{Var}(X) = \dfrac{1}{\lambda^2}$, σ = mean

Median $= \dfrac{\ln 2}{\lambda} \approx 0.693 \times \text{mean}$ (less than the mean, because it is skewed right)

Used for waiting times, lifetimes, and time between events.

⚠ **TRAP "Rata-rata waktu 90 menit" is the MEAN, so λ = 1/90 per minute.** The class board first wrote "λ = rata-rata", then corrected it to E(X) = 1/λ. If you are given a *rate* ("3 per jam", 3 per hour), then λ = 3 and the mean is 1/3 hour.

Link to Poisson: if events occur at rate λ per unit time (a Poisson process), the time T until the next event is Exp(λ): $\text{P}(T > t) = \text{P}(0 \text{ events in } [0, t]) = e^{-\lambda t}$.

Example: 2 calls per hour. P(wait more than 1 hour) = e^{-2} = 0.1353. P(next call within 15 min) = 1 - e^(-2·0.25) = 0.3935.

The time to finish a statistics exam has mean 90 minutes. X ~ Exp(λ = 1/90). class board

1. "dalam waktu 88 menit" (in 88 minutes). The board wrote P(X = 88) = f(88). For a continuous X, **P(X = 88) = 0**, and f(88) = (1/90)e^(-88/90) = 0.00418 is a *density*, not a probability. The sensible reading is "within 88 minutes": P(X ≤ 88) = 1 - e^(-88/90) = **0.6239**. On the exam, write which reading you used. If they insist on f(88), give 0.00418 and add "(density, P(X = 88) = 0)".
2. P(X < 100) = F(100) = 1 - e^(-100/90) = **0.6708**
3. P(X > 75) = 1 - F(75) = e^(-75/90) = **0.4346**

Median time = 90 ln 2 = 62.38 min.

Memoryless property

$$\text{P}(X \geq t + t_0 \mid X \geq t_0) = \text{P}(X \geq t) = e^{-\lambda t}$$

How long it has already lasted does not matter. Only the **additional** time t counts.

Light bulb (Devore p.172 type): mean life 1000 h. P(it lasts another 500 h given it has already lasted 1000 h) = e^(-500/1000) = **0.6065**, the same as for a new bulb.

Exam: P(X > 120 | X > 60) = e^(-60/90) = 0.5134.

Functions of a continuous random variable: Y = g(X)

CDF method (always works):

1. Find the range of Y.
2. $F_Y(y) = \text{P}(Y \leq y) = \text{P}(g(X) \leq y)$. Rewrite this as an event about X and use F_X.
3. $f_Y(y) = F_Y'(y)$.

Monotone g (shortcut): $f_Y(y) = f_X(g^{-1}(y)) \left| \dfrac{d}{dy} g^{-1}(y) \right|$.

Only E(Y) needed? Use LOTUS: $\text{E}[g(X)] = \int g(x) f_X(x)\, dx$. There is no need for f_Y.

Ex. X ~ U(0, 1), Y = e^X. Range [1, e]. $F_Y(y) = \text{P}(X \leq \ln y) = \ln y$, so $f_Y(y) = 1/y$ on [1, e]. E(Y) = $\int_0^1 e^x\, dx$ = e - 1.

Ex. X ~ U(-1, 1), Y = X^2. $F_Y(y) = \text{P}(-\sqrt{y} \leq X \leq \sqrt{y}) = \sqrt{y}$, so $f_Y = \dfrac{1}{2\sqrt{y}}$ on (0, 1].

Linear: Y = aX + b: E = aμ + b, Var = $a^2\sigma^2$. If X is Normal, Y ~ N(aμ + b, $a^2\sigma^2$).

Ex. (reading): X ~ N(2, 4), Y = 3 - 2X ~ N(-1, 16).

X · Traps and errata

Mistakes on the class boards / in your notes (do not copy these)

Where	Board / notes say	Correct
Continuous contoh soal 2	f(x) = 0,75x + 0,2 on [3, 5], then divide by 6.4	Must be **0,075x + 0,2**, which integrates to 1. a) 0.4625, b) 0.5375, c) 0.5, d) 0.2781, μ = 4.05, σ^2 = 0.3308 (**p.19**)
Same, part c (board)	7.594 + 0.9 - 4.594 - 0.7 = 4.6	= 3.2 (arithmetic slip)
Notes Probstat 9, board	"P(X ≤ x) = f(x) = pdf"	P(X ≤ x) = **F(x)** = cdf. The pdf f(x) is the density.
Notes Probstat 5, board	"cdf = cumulative density function"	cumulative **distribution** function
Notes Probstat 5	Var = $E(x^2)$ - $[E(x)^2]$	Var = $E(X^2)$ - $[E(X)]^2$
Binomial B(8, 0.4)	56 · 0.64 · 0.077	0.4^3 = **0.064**. Result 0.2787 ✓
Binomial B(8, 0.4), P(X ≤ 3)	0.592	0.5941 (rounding too early)
Exponential example	"λ = rata-rata"; P(X = 88) = f(88)	mean = 1/λ; P(X = 88) = 0 for continuous X (**p.22**)
N(8.46, 0.84)	σ = 0.913	σ = √0.84 = 0.9165
N(15015, 3540^2) integral	integral = 69.6	≈ 696 (×$1.127 \cdot 10^{-4}$ = 0.078)
Normal ∫ from 0	$\int_0^x$ for P(X ≤ x)	Strictly $\int_{-\infty}^x$. From 0 is fine only when μ is many σ above 0.
Dice-difference pairs	(1,5) under X = 1; (7,6) under X = 3	(6,5) and (3,6). Counts 6, 10, 8, 6, 4, 2 ✓
Sports problem (d)	P(woman ∩ walking) = 57.7/86.4	That is **P(woman | walking)** = 0.6678
Variance sign	"σ^2 and $s^2 > 0$", "s = ±2"	σ^2, $s^2 \geq 0$ and s = +√s^2 only
Z-table limits	Z_{min} = -5.49	The table runs to ±3.49. Beyond that, Φ ≈ 0 or 1.

Exam traps

- **σ vs σ^2** in N(μ, σ^2), and on the calculator (Normal CD needs σ).
- **Discrete ≤ vs <**: P(X < 3) = P(X ≤ 2) for Binomial/Poisson. "Paling sedikit 2" means P(X ≥ 2) = 1 - P(X ≤ 1), **not** 1 - P(X ≤ 2).
- **Poisson λ must match the interval.** 4 per hour gives λ = 2 for 30 minutes and λ = 8 for 2 hours.
- **Exponential mean vs rate**: λ = 1/mean.
- **Without replacement** means Hypergeometric, not Binomial.
- **Independent ≠ mutually exclusive.**
- **Conditional denominator** = the total of the *given* event.
- **P(A | B) ≠ P(B | A)**, so use Bayes to flip.
- **Sample variance divides by n - 1** (sx on the calculator, not σx).
- **Var(aX + b) = a^2Var(X)**. The b disappears and a is squared. Var(X - Y) for independent X, Y = Var X + Var Y.
- **Continuous: P(X = a) = 0**, and f(a) can be > 1.
- **Integrate over the support only.** Split the integral at the support boundaries.
- **Median of skewed data**: right skew means mean > median.
- **Chebyshev gives "at least"** and works for any shape. The empirical rule gives "approximately" and needs a mound shape.
- **Round z to 2 decimals** for the table, but keep full precision elsewhere.
- **Indonesian decimal comma**: 0,4 = 0.4 and 15.015 = 15015.

T.1 Standard Normal Table: Φ(z) = P(Z ≤ z), z ≤ 0

Row = z to one decimal, column = second decimal. For negative z: z = -1.36 → row -1.3, column .06. Area is to the **LEFT** of z *(luas di kiri z)*.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
-3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
-3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
-3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
-3.0	0.0013	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

T.2 Standard Normal Table: Φ(z) = P(Z ≤ z), z ≥ 0

Row = z to one decimal, column = second decimal. z = 1.36 → row 1.3, column .06 → .9131. Area is to the **LEFT** of z *(luas di kiri z)*.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Handy critical values: $z_{.10} = 1.28$, $z_{.05} = 1.645$, $z_{.025} = 1.96$, $z_{.01} = 2.33$, $z_{.005} = 2.58$ (area to the RIGHT). Beyond the table: $z \geq 3.50 \rightarrow \Phi \approx 1.0000$; $z \leq -3.50 \rightarrow \Phi \approx 0.0000$.

T.3 Cumulative Binomial: B(x; n, p) = P(X ≤ x)

P(X = k) = B(k) - B(k-1). P(X ≥ k) = 1 - B(k-1). P(X > k) = 1 - B(k). Calculator check: **Binomial CD** gives the same value.

n = 5 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.7738	.5905	.3277	.2373	.1681	.0778	.0312	.0102	.0024	.0010	.0003	.0000	.0000
1	.9774	.9185	.7373	.6328	.5282	.3370	.1875	.0870	.0308	.0156	.0067	.0005	.0000
2	.9988	.9914	.9421	.8965	.8369	.6826	.5000	.3174	.1631	.1035	.0579	.0086	.0012
3	1.000	.9995	.9933	.9844	.9692	.9130	.8125	.6630	.4718	.3672	.2627	.0815	.0226
4	1.000	1.000	.9997	.9990	.9976	.9898	.9688	.9222	.8319	.7627	.6723	.4095	.2262
5	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 6 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.7351	.5314	.2621	.1780	.1176	.0467	.0156	.0041	.0007	.0002	.0001	.0000	.0000
1	.9672	.8857	.6554	.5339	.4202	.2333	.1094	.0410	.0109	.0046	.0016	.0001	.0000
2	.9978	.9842	.9011	.8306	.7443	.5443	.3438	.1792	.0705	.0376	.0170	.0013	.0001
3	.9999	.9987	.9830	.9624	.9295	.8208	.6562	.4557	.2557	.1694	.0989	.0158	.0022
4	1.000	.9999	.9984	.9954	.9891	.9590	.8906	.7667	.5798	.4661	.3446	.1143	.0328
5	1.000	1.000	.9999	.9998	.9993	.9959	.9844	.9533	.8824	.8220	.7379	.4686	.2649
6	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 7 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.6983	.4783	.2097	.1335	.0824	.0280	.0078	.0016	.0002	.0001	.0000	.0000	.0000
1	.9556	.8503	.5767	.4449	.3294	.1586	.0625	.0188	.0038	.0013	.0004	.0000	.0000
2	.9962	.9743	.8520	.7564	.6471	.4199	.2266	.0963	.0288	.0129	.0047	.0002	.0000
3	.9998	.9973	.9667	.9294	.8740	.7102	.5000	.2898	.1260	.0706	.0333	.0027	.0002
4	1.000	.9998	.9953	.9871	.9712	.9037	.7734	.5801	.3529	.2436	.1480	.0257	.0038
5	1.000	1.000	.9996	.9987	.9962	.9812	.9375	.8414	.6706	.5551	.4233	.1497	.0444
6	1.000	1.000	1.000	.9999	.9998	.9984	.9922	.9720	.9176	.8665	.7903	.5217	.3017
7	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 8 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.6634	.4305	.1678	.1001	.0576	.0168	.0039	.0007	.0001	.0000	.0000	.0000	.0000
1	.9428	.8131	.5033	.3671	.2553	.1064	.0352	.0085	.0013	.0004	.0001	.0000	.0000
2	.9942	.9619	.7969	.6785	.5518	.3154	.1445	.0498	.0113	.0042	.0012	.0000	.0000
3	.9996	.9950	.9437	.8862	.8059	.5941	.3633	.1737	.0580	.0273	.0104	.0004	.0000
4	1.000	.9996	.9896	.9727	.9420	.8263	.6367	.4059	.1941	.1138	.0563	.0050	.0004
5	1.000	1.000	.9988	.9958	.9887	.9502	.8555	.6846	.4482	.3215	.2031	.0381	.0058
6	1.000	1.000	.9999	.9996	.9987	.9915	.9648	.8936	.7447	.6329	.4967	.1869	.0572
7	1.000	1.000	1.000	1.000	.9999	.9993	.9961	.9832	.9424	.8999	.8322	.5695	.3366
8	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 9 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.6302	.3874	.1342	.0751	.0404	.0101	.0020	.0003	.0000	.0000	.0000	.0000	.0000
1	.9288	.7748	.4362	.3003	.1960	.0705	.0195	.0038	.0004	.0001	.0000	.0000	.0000
2	.9916	.9470	.7382	.6007	.4628	.2318	.0898	.0250	.0043	.0013	.0003	.0000	.0000
3	.9994	.9917	.9144	.8343	.7297	.4826	.2539	.0994	.0253	.0100	.0031	.0001	.0000
4	1.000	.9991	.9804	.9511	.9012	.7334	.5000	.2666	.0988	.0489	.0196	.0009	.0000
5	1.000	.9999	.9969	.9900	.9747	.9006	.7461	.5174	.2703	.1657	.0856	.0083	.0006
6	1.000	1.000	.9997	.9987	.9957	.9750	.9102	.7682	.5372	.3993	.2618	.0530	.0084
7	1.000	1.000	1.000	.9999	.9996	.9962	.9805	.9295	.8040	.6997	.5638	.2252	.0712
8	1.000	1.000	1.000	1.000	1.000	.9997	.9980	.9899	.9596	.9249	.8658	.6126	.3698
9	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

T.3 Cumulative Binomial: B(x; n, p) = P(X ≤ x) (cont.)

n = 10 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.5987	.3487	.1074	.0563	.0282	.0060	.0010	.0001	.0000	.0000	.0000	.0000	.0000
1	.9139	.7361	.3758	.2440	.1493	.0464	.0107	.0017	.0001	.0000	.0000	.0000	.0000
2	.9885	.9298	.6778	.5256	.3828	.1673	.0547	.0123	.0016	.0004	.0001	.0000	.0000
3	.9990	.9872	.8791	.7759	.6496	.3823	.1719	.0548	.0106	.0035	.0009	.0000	.0000
4	.9999	.9984	.9672	.9219	.8497	.6331	.3770	.1662	.0473	.0197	.0064	.0001	.0000
5	1.000	.9999	.9936	.9803	.9527	.8338	.6230	.3669	.1503	.0781	.0328	.0016	.0001
6	1.000	1.000	.9991	.9965	.9894	.9452	.8281	.6177	.3504	.2241	.1209	.0128	.0010
7	1.000	1.000	.9999	.9996	.9984	.9877	.9453	.8327	.6172	.4744	.3222	.0702	.0115
8	1.000	1.000	1.000	1.000	.9999	.9983	.9893	.9536	.8507	.7560	.6242	.2639	.0861
9	1.000	1.000	1.000	1.000	1.000	.9999	.9990	.9940	.9718	.9437	.8926	.6513	.4013
10	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 12 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.5404	.2824	.0687	.0317	.0138	.0022	.0002	.0000	.0000	.0000	.0000	.0000	.0000
1	.8816	.6590	.2749	.1584	.0850	.0196	.0032	.0003	.0000	.0000	.0000	.0000	.0000
2	.9804	.8891	.5583	.3907	.2528	.0834	.0193	.0028	.0002	.0000	.0000	.0000	.0000
3	.9978	.9744	.7946	.6488	.4925	.2253	.0730	.0153	.0017	.0004	.0001	.0000	.0000
4	.9998	.9957	.9274	.8424	.7237	.4382	.1938	.0573	.0095	.0028	.0006	.0000	.0000
5	1.000	.9995	.9806	.9456	.8822	.6652	.3872	.1582	.0386	.0143	.0039	.0001	.0000
6	1.000	.9999	.9961	.9857	.9614	.8418	.6128	.3348	.1178	.0544	.0194	.0005	.0000
7	1.000	1.000	.9994	.9972	.9905	.9427	.8062	.5618	.2763	.1576	.0726	.0043	.0002
8	1.000	1.000	.9999	.9996	.9983	.9847	.9270	.7747	.5075	.3512	.2054	.0256	.0022
9	1.000	1.000	1.000	1.000	.9998	.9972	.9807	.9166	.7472	.6093	.4417	.1109	.0196
10	1.000	1.000	1.000	1.000	1.000	.9997	.9968	.9804	.9150	.8416	.7251	.3410	.1184
11	1.000	1.000	1.000	1.000	1.000	1.000	.9998	.9978	.9862	.9683	.9313	.7176	.4596
12	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 15 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.4633	.2059	.0352	.0134	.0047	.0005	.0000	.0000	.0000	.0000	.0000	.0000	.0000
1	.8290	.5490	.1671	.0802	.0353	.0052	.0005	.0000	.0000	.0000	.0000	.0000	.0000
2	.9638	.8159	.3980	.2361	.1268	.0271	.0037	.0003	.0000	.0000	.0000	.0000	.0000
3	.9945	.9444	.6482	.4613	.2969	.0905	.0176	.0019	.0001	.0000	.0000	.0000	.0000
4	.9994	.9873	.8358	.6865	.5155	.2173	.0592	.0093	.0007	.0001	.0000	.0000	.0000
5	.9999	.9978	.9389	.8516	.7216	.4032	.1509	.0338	.0037	.0008	.0001	.0000	.0000
6	1.000	.9997	.9819	.9434	.8689	.6098	.3036	.0950	.0152	.0042	.0008	.0000	.0000
7	1.000	1.000	.9958	.9827	.9500	.7869	.5000	.2131	.0500	.0173	.0042	.0000	.0000
8	1.000	1.000	.9992	.9958	.9848	.9050	.6964	.3902	.1311	.0566	.0181	.0003	.0000
9	1.000	1.000	.9999	.9992	.9963	.9662	.8491	.5968	.2784	.1484	.0611	.0022	.0001
10	1.000	1.000	1.000	.9999	.9993	.9907	.9408	.7827	.4845	.3135	.1642	.0127	.0006
11	1.000	1.000	1.000	1.000	.9999	.9981	.9824	.9095	.7031	.5387	.3518	.0556	.0055
12	1.000	1.000	1.000	1.000	1.000	.9997	.9963	.9729	.8732	.7639	.6020	.1841	.0362
13	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9948	.9647	.9198	.8329	.4510	.1710
14	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9953	.9866	.9648	.7941	.5367
15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

T.3 Cumulative Binomial: B(x; n, p) = P(X ≤ x) (cont.)

n = 20 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.3585	.1216	.0115	.0032	.0008	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
1	.7358	.3917	.0692	.0243	.0076	.0005	.0000	.0000	.0000	.0000	.0000	.0000	.0000
2	.9245	.6769	.2061	.0913	.0355	.0036	.0002	.0000	.0000	.0000	.0000	.0000	.0000
3	.9841	.8670	.4114	.2252	.1071	.0160	.0013	.0000	.0000	.0000	.0000	.0000	.0000
4	.9974	.9568	.6296	.4148	.2375	.0510	.0059	.0003	.0000	.0000	.0000	.0000	.0000
5	.9997	.9887	.8042	.6172	.4164	.1256	.0207	.0016	.0000	.0000	.0000	.0000	.0000
6	1.000	.9976	.9133	.7858	.6080	.2500	.0577	.0065	.0003	.0000	.0000	.0000	.0000
7	1.000	.9996	.9679	.8982	.7723	.4159	.1316	.0210	.0013	.0002	.0000	.0000	.0000
8	1.000	.9999	.9900	.9591	.8867	.5956	.2517	.0565	.0051	.0009	.0001	.0000	.0000
9	1.000	1.000	.9974	.9861	.9520	.7553	.4119	.1275	.0171	.0039	.0006	.0000	.0000
10	1.000	1.000	.9994	.9961	.9829	.8725	.5881	.2447	.0480	.0139	.0026	.0000	.0000
11	1.000	1.000	.9999	.9991	.9949	.9435	.7483	.4044	.1133	.0409	.0100	.0001	.0000
12	1.000	1.000	1.000	.9998	.9987	.9790	.8684	.5841	.2277	.1018	.0321	.0004	.0000
13	1.000	1.000	1.000	1.000	.9997	.9935	.9423	.7500	.3920	.2142	.0867	.0024	.0000
14	1.000	1.000	1.000	1.000	1.000	.9984	.9793	.8744	.5836	.3828	.1958	.0113	.0003
15	1.000	1.000	1.000	1.000	1.000	.9997	.9941	.9490	.7625	.5852	.3704	.0432	.0026
16	1.000	1.000	1.000	1.000	1.000	1.000	.9987	.9840	.8929	.7748	.5886	.1330	.0159
17	1.000	1.000	1.000	1.000	1.000	1.000	.9998	.9964	.9645	.9087	.7939	.3231	.0755
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9924	.9757	.9308	.6083	.2642
19	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9992	.9968	.9885	.8784	.6415
20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

n = 25 x	.05	.10	.20	.25	.30	.40	.50	.60	.70	.75	.80	.90	.95
0	.2774	.0718	.0038	.0008	.0001	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
1	.6424	.2712	.0274	.0070	.0016	.0001	.0000	.0000	.0000	.0000	.0000	.0000	.0000
2	.8729	.5371	.0982	.0321	.0090	.0004	.0000	.0000	.0000	.0000	.0000	.0000	.0000
3	.9659	.7636	.2340	.0962	.0332	.0024	.0001	.0000	.0000	.0000	.0000	.0000	.0000
4	.9928	.9020	.4207	.2137	.0905	.0095	.0005	.0000	.0000	.0000	.0000	.0000	.0000
5	.9988	.9666	.6167	.3783	.1935	.0294	.0020	.0001	.0000	.0000	.0000	.0000	.0000
6	.9998	.9905	.7800	.5611	.3407	.0736	.0073	.0003	.0000	.0000	.0000	.0000	.0000
7	1.000	.9977	.8909	.7265	.5118	.1536	.0216	.0012	.0000	.0000	.0000	.0000	.0000
8	1.000	.9995	.9532	.8506	.6769	.2735	.0539	.0043	.0001	.0000	.0000	.0000	.0000
9	1.000	.9999	.9827	.9287	.8106	.4246	.1148	.0132	.0005	.0000	.0000	.0000	.0000
10	1.000	1.000	.9944	.9703	.9022	.5858	.2122	.0344	.0018	.0002	.0000	.0000	.0000
11	1.000	1.000	.9985	.9893	.9558	.7323	.3450	.0778	.0060	.0009	.0001	.0000	.0000
12	1.000	1.000	.9996	.9966	.9825	.8462	.5000	.1538	.0175	.0034	.0004	.0000	.0000
13	1.000	1.000	.9999	.9991	.9940	.9222	.6550	.2677	.0442	.0107	.0015	.0000	.0000
14	1.000	1.000	1.000	.9998	.9982	.9656	.7878	.4142	.0978	.0297	.0056	.0000	.0000
15	1.000	1.000	1.000	1.000	.9995	.9868	.8852	.5754	.1894	.0713	.0173	.0001	.0000
16	1.000	1.000	1.000	1.000	.9999	.9957	.9461	.7265	.3231	.1494	.0468	.0005	.0000
17	1.000	1.000	1.000	1.000	1.000	.9988	.9784	.8464	.4882	.2735	.1091	.0023	.0000
18	1.000	1.000	1.000	1.000	1.000	.9997	.9927	.9264	.6593	.4389	.2200	.0095	.0002
19	1.000	1.000	1.000	1.000	1.000	.9999	.9980	.9706	.8065	.6217	.3833	.0334	.0012
20	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9905	.9095	.7863	.5793	.0980	.0072
21	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9976	.9668	.9038	.7660	.2364	.0341
22	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9996	.9910	.9679	.9018	.4629	.1271
23	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9984	.9930	.9726	.7288	.3576
24	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9992	.9962	.9282	.7226
25	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

T.4 Cumulative Poisson: F(x; λ) = P(X ≤ x)

λ = mean number of events in the interval (scale it to the interval asked: λ_{new} = rate × length). P(X = k) = F(k) - F(k-1). Calculator check: **Poisson CD**.

x \ λ	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1
0	.9048	.8187	.7408	.6703	.6065	.5488	.4966	.4493	.4066	.3679
1	.9953	.9825	.9631	.9384	.9098	.8781	.8442	.8088	.7725	.7358
2	.9998	.9989	.9964	.9921	.9856	.9769	.9659	.9526	.9371	.9197
3	1.000	.9999	.9997	.9992	.9982	.9966	.9942	.9909	.9865	.9810
4	1.000	1.000	1.000	.9999	.9998	.9996	.9992	.9986	.9977	.9963
5	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9998	.9997	.9994
6	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999
7	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
8	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

x \ λ	1.5	2	2.5	3	3.5	4	4.5	5	6	7
0	.2231	.1353	.0821	.0498	.0302	.0183	.0111	.0067	.0025	.0009
1	.5578	.4060	.2873	.1991	.1359	.0916	.0611	.0404	.0174	.0073
2	.8088	.6767	.5438	.4232	.3208	.2381	.1736	.1247	.0620	.0296
3	.9344	.8571	.7576	.6472	.5366	.4335	.3423	.2650	.1512	.0818
4	.9814	.9473	.8912	.8153	.7254	.6288	.5321	.4405	.2851	.1730
5	.9955	.9834	.9580	.9161	.8576	.7851	.7029	.6160	.4457	.3007
6	.9991	.9955	.9858	.9665	.9347	.8893	.8311	.7622	.6063	.4497
7	.9998	.9989	.9958	.9881	.9733	.9489	.9134	.8666	.7440	.5987
8	1.000	.9998	.9989	.9962	.9901	.9786	.9597	.9319	.8472	.7291
9	1.000	1.000	.9997	.9989	.9967	.9919	.9829	.9682	.9161	.8305
10	1.000	1.000	.9999	.9997	.9990	.9972	.9933	.9863	.9574	.9015
11	1.000	1.000	1.000	.9999	.9997	.9991	.9976	.9945	.9799	.9467
12	1.000	1.000	1.000	1.000	.9999	.9997	.9992	.9980	.9912	.9730
13	1.000	1.000	1.000	1.000	1.000	.9999	.9997	.9993	.9964	.9872
14	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9998	.9986	.9943
15	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9995	.9976
16	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9998	.9990
17	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9996
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999

T.4 Cumulative Poisson (cont.)

x \ λ	8	9	10	11	12	13	14	15	18	20
0	.0003	.0001	.0000	.0000	.0000	.0000	.0000	.0000	.0000	.0000
1	.0030	.0012	.0005	.0002	.0001	.0000	.0000	.0000	.0000	.0000
2	.0138	.0062	.0028	.0012	.0005	.0002	.0001	.0000	.0000	.0000
3	.0424	.0212	.0103	.0049	.0023	.0011	.0005	.0002	.0000	.0000
4	.0996	.0550	.0293	.0151	.0076	.0037	.0018	.0009	.0001	.0000
5	.1912	.1157	.0671	.0375	.0203	.0107	.0055	.0028	.0003	.0001
6	.3134	.2068	.1301	.0786	.0458	.0259	.0142	.0076	.0010	.0003
7	.4530	.3239	.2202	.1432	.0895	.0540	.0316	.0180	.0029	.0008
8	.5925	.4557	.3328	.2320	.1550	.0998	.0621	.0374	.0071	.0021
9	.7166	.5874	.4579	.3405	.2424	.1658	.1094	.0699	.0154	.0050
10	.8159	.7060	.5830	.4599	.3472	.2517	.1757	.1185	.0304	.0108
11	.8881	.8030	.6968	.5793	.4616	.3532	.2600	.1848	.0549	.0214
12	.9362	.8758	.7916	.6887	.5760	.4631	.3585	.2676	.0917	.0390
13	.9658	.9261	.8645	.7813	.6815	.5730	.4644	.3632	.1426	.0661
14	.9827	.9585	.9165	.8540	.7720	.6751	.5704	.4657	.2081	.1049
15	.9918	.9780	.9513	.9074	.8444	.7636	.6694	.5681	.2867	.1565
16	.9963	.9889	.9730	.9441	.8987	.8355	.7559	.6641	.3751	.2211
17	.9984	.9947	.9857	.9678	.9370	.8905	.8272	.7489	.4686	.2970
18	.9993	.9976	.9928	.9823	.9626	.9302	.8826	.8195	.5622	.3814
19	.9997	.9989	.9965	.9907	.9787	.9573	.9235	.8752	.6509	.4703
20	.9999	.9996	.9984	.9953	.9884	.9750	.9521	.9170	.7307	.5591
21	1.000	.9998	.9993	.9977	.9939	.9859	.9712	.9469	.7991	.6437
22	1.000	.9999	.9997	.9990	.9970	.9924	.9833	.9673	.8551	.7206
23	1.000	1.000	.9999	.9995	.9985	.9960	.9907	.9805	.8989	.7875
24	1.000	1.000	1.000	.9998	.9993	.9980	.9950	.9888	.9317	.8432
25	1.000	1.000	1.000	.9999	.9997	.9990	.9974	.9938	.9554	.8878
26	1.000	1.000	1.000	1.000	.9999	.9995	.9987	.9967	.9718	.9221
27	1.000	1.000	1.000	1.000	.9999	.9998	.9994	.9983	.9827	.9475
28	1.000	1.000	1.000	1.000	1.000	.9999	.9997	.9991	.9897	.9657
29	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9996	.9941	.9782
30	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9998	.9967	.9865
31	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9982	.9919
32	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9990	.9953
33	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9995	.9973
34	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9998	.9985
35	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9992
36	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.9999	.9996

Q · Practice quiz: Probabilitas & Statistika

Rules: open binder, calculator allowed, **do not look at Tab K until you finish**. Suggested time is 100 minutes for all 16 questions, or about 50 minutes for any 8. Topics are mixed on purpose, so part of the task is deciding which method to use (Tab I and Tab D). Write X = ..., the distribution, the formula, the substitution and the answer to 4 decimals. Each question is in Indonesian with an English line underneath.

1. Data kelulusan 200 mahasiswa mata kuliah Probstat berdasarkan gender: [8]

	Lulus (L)	Tidak Lulus (TL)
Perempuan (F)	72	18
Laki-laki (M)	78	32

Jika seorang mahasiswa dipilih secara acak, hitunglah: (a) P(F ∩ L) (b) P(L) (c) P(L | F) (d) P(F | TL) (e) P(F ∪ L). (f) Apakah kejadian F dan L saling bebas? Jelaskan.

200 students by gender and pass/fail. Pick one at random. Find (a)–(e). (f) Are F and L independent? Explain.

2. Berat bersih kemasan kopi berdistribusi normal dengan rata-rata 500 gram dan varians 25 gram². Tentukan peluang sebuah kemasan memiliki berat: (a) kurang dari 490 gram (b) antara 495 dan 510 gram (c) lebih dari 512 gram. (d) Tentukan berat minimum agar sebuah kemasan termasuk 10% kemasan terberat. [7]

Net weight ~ Normal, mean 500 g, variance 25. Find (a) P(X < 490), (b) P(495 < X < 510), (c) P(X > 512), (d) the weight that marks the heaviest 10%.

3. Distribusi peluang variabel random X diberikan sebagai berikut: [7]

x	0	1	2	3
P(X = x)	0,1	k	2k	0,3

(a) Tentukan nilai k. (b) Hitung P(X ≥ 2) dan F(1,5). (c) Hitung E(X) dan Var(X). (d) Jika Y = 3X - 2, hitung E(Y) dan Var(Y).

(a) Find k. (b) P(X ≥ 2) and F(1.5). (c) E(X), Var(X). (d) For Y = 3X - 2, find E(Y) and Var(Y).

4. Waktu tunggu (menit) 12 pelanggan di sebuah kafe: 4, 7, 5, 9, 6, 12, 5, 8, 30, 6, 7, 5. Hitunglah: (a) rata-rata, median, dan modus (b) varians dan simpangan baku sampel (c) Q1, Q3, dan IQR (d) Apakah ada pencilan? Gunakan batas Q1 - 1,5·IQR dan Q3 + 1,5·IQR. (e) Bagaimana bentuk distribusinya, dan ukuran pemusatan mana yang lebih tepat? [8]

Waiting times of 12 customers. (a) Mean, median, mode. (b) Sample variance and SD. (c) Q1, Q3, IQR. (d) Any outliers (1.5·IQR fences)? (e) Shape, and which measure of centre is better?

5. Rata-rata terdapat 4 panggilan masuk per jam ke layanan helpdesk kampus. Hitunglah peluang: (a) tepat 2 panggilan dalam 1 jam (b) paling banyak 3 panggilan dalam 1 jam (c) paling sedikit 1 panggilan dalam 30 menit (d) lebih dari 5 panggilan dalam 1 jam. [6]

On average 4 calls per hour. (a) Exactly 2 in an hour. (b) At most 3 in an hour. (c) At least 1 in 30 minutes. (d) More than 5 in an hour.

6. Diketahui fungsi $f(x) = \begin{cases} c(x+1) & 1 \le x \le 3 \\ 0 & x \text{ lainnya} \end{cases}$. (a) Tentukan c agar f(x) merupakan pdf. (b) Hitung P(X ≤ 2) dan P(1,5 < X < 2,5). (c) Tentukan fungsi kumulatif F(x). (d) Hitung μ dan σ². (e) Tentukan median X. [8]

(a) Find c so f is a pdf. (b) P(X ≤ 2), P(1.5 < X < 2.5). (c) The cdf F(x). (d) μ and σ². (e) The median.

7. Sebuah pabrik memiliki tiga mesin A, B, dan C yang masing-masing menghasilkan 50%, 30%, dan 20% dari total produksi. Persentase produk cacat dari mesin A, B, dan C berturut-turut adalah 2%, 3%, dan 5%. (a) Berapa peluang sebuah produk yang diambil secara acak cacat? (b) Jika produk yang diambil ternyata cacat, berapa peluang produk tersebut berasal dari mesin C? (c) Mesin mana yang paling mungkin menghasilkan produk cacat tersebut? [7]

Machines A, B, C make 50%, 30%, 20% of output, with defect rates 2%, 3%, 5%. (a) P(defective). (b) P(C | defective). (c) Which machine most likely produced a given defective item?

8. Peluang sebuah komponen elektronik cacat adalah 0,1. Diambil 12 komponen secara acak (saling bebas). Hitunglah peluang: (a) tepat 2 komponen cacat (b) paling banyak 1 komponen cacat (c) paling sedikit 3 komponen cacat (d) sedikitnya satu komponen cacat. (e) Hitung rata-rata dan simpangan baku banyaknya komponen cacat. [6]

P(defective) = 0.1, 12 independent components. (a) Exactly 2. (b) At most 1. (c) At least 3. (d) At least one. (e) Mean and SD.

9. Umur sebuah lampu LED berdistribusi eksponensial dengan rata-rata 1000 jam. Hitunglah: (a) peluang lampu mati sebelum 500 jam (b) peluang lampu bertahan lebih dari 1500 jam (c) peluang lampu bertahan lebih dari 1500 jam jika diketahui lampu sudah menyala 1000 jam (d) median umur lampu. [6]

LED lifetime ~ Exponential, mean 1000 h. (a) P(fails before 500 h). (b) P(lasts more than 1500 h). (c) P(lasts more than 1500 h | it has already lasted 1000 h). (d) Median lifetime.

10. (a) Dari 6 pria dan 4 wanita akan dibentuk panitia beranggotakan 4 orang. Berapa banyak cara membentuk panitia? Berapa peluang panitia terdiri dari tepat 2 wanita? Berapa peluang paling sedikit 1 wanita? (b) Dari 5 kandidat akan dipilih ketua, sekretaris, dan bendahara. Ada berapa cara? [6]

(a) A committee of 4 from 6 men and 4 women: number of ways, P(exactly 2 women), P(at least 1 woman). (b) Choose a chair, secretary and treasurer from 5 candidates: number of ways.

11. Sebuah lab memiliki 15 laptop, 4 di antaranya rusak. Seorang asisten mengambil 5 laptop secara acak **tanpa pengembalian**. Misalkan X = banyaknya laptop rusak yang terambil. (a) Hitung P(X = 1). (b) Hitung peluang paling sedikit satu laptop rusak terambil. (c) Hitung E(X) dan Var(X). [6]

15 laptops, 4 broken. Pick 5 without replacement; X = number broken. (a) P(X = 1). (b) P(at least one broken). (c) E(X), Var(X).

12. Buktikan bahwa fungsi $f(x) = \begin{cases} 0{,}2\,e^{-0{,}2(x-1)} & x \geq 1 \\ 0 & x \text{ lainnya} \end{cases}$ merupakan sebuah pdf. Kemudian hitung P(X > 6) dan P(2 < X < 4). [6]

Prove that f is a pdf. Then find P(X > 6) and P(2 < X < 4).

13. Nilai UTS suatu kelas memiliki rata-rata 70 dan simpangan baku 8. (a) Tanpa mengetahui bentuk distribusinya, paling sedikit berapa proporsi mahasiswa yang nilainya antara 54 dan 86? (b) Jika distribusinya berbentuk lonceng (mound-shaped), kira-kira berapa proporsi mahasiswa yang nilainya di atas 86? (c) ... dan antara 62 dan 86? [5]

Midterm scores: mean 70, SD 8. (a) With no shape assumption, at least what proportion is between 54 and 86? (b) If mound-shaped, roughly what proportion is above 86? (c) Between 62 and 86?

14. Diketahui 30% pengguna aplikasi membuka fitur premium. Diambil sampel 100 pengguna. Dengan pendekatan distribusi normal, hitunglah peluang (a) paling banyak 25 pengguna membuka fitur premium (b) tepat 30 pengguna. (c) Bandingkan dengan nilai eksak Binomial (gunakan kalkulator). [6]

30% use the premium feature, n = 100. Using the normal approximation: (a) P(at most 25). (b) P(exactly 30). (c) Compare with the exact Binomial.

15. Bus kampus datang setiap 20 menit. Waktu tunggu seorang mahasiswa yang datang secara acak berdistribusi seragam pada [0, 20] menit. Hitung (a) peluang menunggu lebih dari 15 menit (b) peluang menunggu antara 5 dan 12 menit (c) rata-rata dan varians waktu tunggu. [4]

Wait ~ Uniform[0, 20] min. (a) P(W > 15). (b) P(5 < W < 12). (c) Mean and variance.

16. Sebuah gedung memiliki 3 alarm asap yang bekerja saling bebas. Masing-masing mendeteksi asap dengan peluang 0,85. (a) Berapa peluang paling sedikit satu alarm berbunyi saat terjadi kebakaran? (b) Berapa peluang tepat dua alarm berbunyi? (c) Apakah kejadian "alarm 1 berbunyi" dan "alarm 2 berbunyi" saling lepas? Jelaskan. [4]

3 independent smoke alarms, each detects with probability 0.85. (a) P(at least one sounds). (b) P(exactly two). (c) Are "alarm 1 sounds" and "alarm 2 sounds" mutually exclusive? Explain.

— Selesai (end) — Total 100 points. Now mark your work with Tab K.

K · Answer key: worked solutions

The "→ p." links point to the binder page with the method. Answers are given to 4 decimals. A z-table answer may differ from Normal CD in the 3rd–4th decimal, and both are accepted.

1. Contingency table → p.13

Totals: F = 90, M = 110, L = 150, TL = 50, N = 200.

(a) P(F ∩ L) = 72/200 = **0.36** (b) P(L) = 150/200 = **0.75** (c) P(L | F) = 72/90 = **0.80** (d) P(F | TL) = 18/50 = **0.36**

(e) P(F ∪ L) = (90 + 150 - 72)/200 = 168/200 = **0.84**

(f) P(L | F) = 0.80 ≠ P(L) = 0.75. Equivalently, P(F)P(L) = 0.45 × 0.75 = 0.3375 ≠ 0.36. So F and L are **not independent**: female students pass at a higher rate.

2. Normal → p.20, p.21

X ~ N(500, 25), so **σ = 5** (not 25!).

(a) z = (490 - 500)/5 = -2.00 → Φ(-2.00) = **0.0228**

(b) z = -1.00 and 2.00 → 0.9772 - 0.1587 = **0.8185**

(c) z = 12/5 = 2.40 → 1 - 0.9918 = **0.0082**

(d) Heaviest 10% means a left area of 0.90, so z = 1.28 and x = 500 + 1.28(5) = **506.4 g** (Inverse Normal: 506.41)

3. Discrete table → p.15, p.15

(a) 0.1 + k + 2k + 0.3 = 1 → **k = 0.2**, so the table is 0.1, 0.2, 0.4, 0.3.

(b) P(X ≥ 2) = 0.4 + 0.3 = **0.7**; F(1.5) = P(X ≤ 1.5) = P(0) + P(1) = **0.3**

(c) E(X) = 0 + 0.2 + 0.8 + 0.9 = **1.9**; $E(X^2)$ = 0 + 0.2 + 1.6 + 2.7 = 4.5; Var(X) = 4.5 - 1.9^2 = **0.89**

(d) E(Y) = 3(1.9) - 2 = **3.7**; Var(Y) = 3^2(0.89) = **8.01** (the -2 does not change the variance)

4. Descriptive → p.10, p.11

Sorted: 4 5 5 5 6 6 7 7 8 9 12 30 (n = 12), Σx = 104, Σx^2 = 1450.

(a) x̄ = 104/12 = **8.6667**; median position 6.5 → (6 + 7)/2 = **6.5**; mode = **5**

(b) $s^2 = \frac{1450 - 104^2/12}{11} = \frac{548.667}{11} =$ **49.8788**, s = **7.0625** (STAT: sx)

(c) Q1 position 0.25(13) = 3.25 → 5 + 0.25(5 - 5) = **5**; Q3 position 9.75 → 8 + 0.75(9 - 8) = **8.75**; IQR = **3.75**

(d) LL = 5 - 5.625 = -0.625, UL = 8.75 + 5.625 = 14.375, so **30 is an outlier**

(e) Mean > median, so the distribution is **skewed right** because of the outlier. The **median** is the better measure of centre: without the 30, x̄ = 6.73 but the median barely moves, to 6.

5. Poisson → p.16

X = number of calls per hour, X ~ Poisson(4).

(a) P(X = 2) = $e^{-4}4^2/2!$ = **0.1465**

(b) P(X ≤ 3) = e^{-4}(1 + 4 + 8 + 10.667) = 0.0183 + 0.0733 + 0.1465 + 0.1954 = **0.4335**

(c) 30 minutes, so **λ = 2**: P(Y ≥ 1) = 1 - e^{-2} = **0.8647**

(d) P(X > 5) = 1 - P(X ≤ 5) = 1 - 0.7851 = **0.2149** (Poisson CD x = 5, μ = 4)

6. General pdf → p.18, p.18, p.18

(a) $\int_1^3 c(x+1)dx = c\left[\frac{x^2}{2} + x\right]_1^3 = c(7.5 - 1.5) = 6c = 1 \Rightarrow$ **c = 1/6**. f ≥ 0 on [1, 3] ✓

(b) $\mathrm{P}(X \le 2) = \frac{1}{6}\left[\frac{x^2}{2} + x\right]_1^2 = \frac{1}{6}(4 - 1.5) =$ **0.4167**; P(1.5 < X < 2.5) = F(2.5) - F(1.5) = 0.6875 - 0.1875 = **0.5**

(c) $F(x) = \begin{cases} 0 & x < 1 \\ \frac{x^2+2x-3}{12} & 1 \le x \le 3 \\ 1 & x > 3 \end{cases}$. Check: F(1) = 0 and F(3) = 12/12 = 1 ✓

(d) $\mu = \frac{1}{6}\int_1^3 (x^2 + x)dx = \frac{1}{6}\left[\frac{x^3}{3} + \frac{x^2}{2}\right]_1^3 = \frac{1}{6}\left(13.5 - \frac{5}{6}\right) = \frac{19}{9} =$ **2.1111**

$\mathrm{E}(X^2) = \frac{1}{6}\left[\frac{x^4}{4} + \frac{x^3}{3}\right]_1^3 = \frac{1}{6}\left(29.25 - \frac{7}{12}\right) = \frac{43}{9} = 4.7778$; σ^2 = 43/9 - (19/9)² = 26/81 = **0.3210**

(e) F(m) = 0.5 → m^2 + 2m - 3 = 6 → m^2 + 2m - 9 = 0 → m = -1 + √10 = **2.1623** (take the root inside [1, 3])

7. Total probability + Bayes → p.14

(a) P(D) = 0.5(0.02) + 0.3(0.03) + 0.2(0.05) = 0.010 + 0.009 + 0.010 = **0.029**

(b) P(C | D) = 0.010/0.029 = **0.3448**

(c) P(A | D) = 0.3448, P(B | D) = 0.3103, P(C | D) = 0.3448. **A and C are tied** as the most likely source. A has the lowest defect rate but makes the most products. Check: the three add to 1.

8. Binomial → p.16

X = number of defective components, X ~ B(12, 0.1).

(a) $\binom{12}{2}(0.1)^2(0.9)^{10} = 66(0.01)(0.3487) =$ **0.2301**

(b) P(0) + P(1) = 0.2824 + 0.3766 = **0.6590**

(c) P(X ≥ 3) = 1 - P(X ≤ 2) = 1 - (0.6590 + 0.2301) = **0.1109**

(d) 1 - P(0) = 1 - 0.9^{12} = **0.7176**

(e) μ = 12(0.1) = **1.2** ; σ = √(12 · 0.1 · 0.9) = √1.08 = **1.0392**

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9. Exponential → p.22, p.22

Mean 1000, so λ = 1/1000 = 0.001 per hour.

(a) P(X < 500) = 1 - e^(-0.5) = **0.3935** (b) P(X > 1500) = e^(-1.5) = **0.2231**

(c) Memoryless: P(X > 1500 | X > 1000) = P(X > 500) = e^(-0.5) = **0.6065**

(d) Median = ln 2/λ = **693.15 h**

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10. Counting → p.12

(a) Ways = C(10, 4) = **210** . P(exactly 2 women) = C(4,2)·C(6,2)/210 = 6·15/210 = **0.4286** . P(at least 1 woman) = 1 - C(6,4)/210 = 1 - 15/210 = **0.9286**

(b) The roles are different, so order matters: P(5, 3) = 5·4·3 = **60**

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11. Hypergeometric → p.17

N = 15, M = 4 broken, n = 5.

(a) $P(X=1) = \frac{\binom{4}{1}\binom{11}{4}}{\binom{15}{5}} = \frac{4 \cdot 330}{3003} =$ **0.4396**

(b) 1 - P(X = 0) = 1 - C(11,5)/C(15,5) = 1 - 462/3003 = **0.8462**

(c) E(X) = 5(4/15) = **1.3333** ; Var(X) = (10/14)(5)(4/15)(11/15) = **0.6984**

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12. Prove pdf → p.18

(1) For x ≥ 1, 0.2 > 0 and e^(·) > 0, so f(x) > 0. Elsewhere f = 0. So f ≥ 0 ✓

(2) $\int_{-\infty}^{\infty} f = \int_1^{\infty} 0.2e^{-0.2(x-1)}dx = \left[-e^{-0.2(x-1)}\right]_1^{\infty} = 0-(-1) = 1$ ✓ ∴ f is a pdf.

P(X > 6) = $\left[-e^{-0.2(x-1)}\right]_6^{\infty} = e^{-1} =$ **0.3679** ; P(2 < X < 4) = e^(-0.2) - e^(-0.6) = 0.8187 - 0.5488 = **0.2699**

This is an exponential distribution with λ = 0.2 shifted to start at 1, so E(X) = 1 + 5 = 6.

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13. Chebyshev and empirical rule → p.10

(a) 54 and 86 are 70 ± 16 = 70 ± 2(8), so k = 2: at least 1 - 1/4 = **0.75**

(b) 86 = μ + 2σ, so ≈ (1 - 0.95)/2 = **0.025**

(c) 62 = μ - σ, 86 = μ + 2σ: ≈ 0.68/2 + 0.95/2 = **0.815** (the z-table gives 0.8185)

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14. Normal approximation → p.21

X ~ B(100, 0.3). np = 30 ≥ 5 and nq = 70 ≥ 5, so the approximation is OK. μ = 30, σ = √21 = 4.5826.

(a) P(X ≤ 25) ≈ P(Y ≤ 25.5): z = (25.5 - 30)/4.5826 = -0.98 → **0.1635**

(b) P(X = 30) ≈ P(29.5 ≤ Y ≤ 30.5): z = ±0.11 → 0.5438 - 0.4562 = **0.0876**

(c) Exact: Binomial CD(25; 100, 0.3) = **0.1631**, Binomial PD(30) = **0.0868**. The approximation is very close.

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15. Uniform → p.19

(a) (20 - 15)/20 = **0.25** (b) (12 - 5)/20 = **0.35** (c) μ = 10 min, $\sigma^2 = 20^2/12 =$ **33.33**

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16. Independence → p.12, p.14

(a) 1 - P(none) = 1 - 0.15^3 = **0.9966**

(b) B(3, 0.85): $3(0.85)^2(0.15) =$ **0.3251**

(c) **No.** They can happen together: P(both) = 0.85^2 = 0.7225 ≠ 0. The alarms are independent but not mutually exclusive.

Self-score

Score	What to do next
Missed any σ vs σ^2, ≤ vs <, or λ-rescaling	Re-read Tab X before the exam. These are exactly the class-board traps.
Lost time choosing a method	Use Tab I first next time, before reading the chapter pages.
Calculation slips	Keep unrounded values in STO memory and run the DIST check values (Tab C) tonight.